

NMAM Institute of Technology, Nitte

Off-Campus Centre of Nitte (Deemed to be University)

COMPLETE EXAMINATION QUESTION BANK

Unified Edition with Single Navigation Menu

IV Semester B.E./B.Tech (CBCS)

Academic Years: 2022–2025

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COMPLETE EXAMINATION QUESTION BANK

IV Semester B.E./B.Tech (CBCS)

Academic Years: 2022–2025

Subjects (Click to Navigate):

Code	Subject
MA2005-1	Linear Algebra and Numerical Methods
CS2102-1	Database Management Systems
CS3004-1	Design and Analysis of Algorithms
CS2103-1	Software Engineering & Project Management
CS3005-1	Microprocessor and Embedded Systems
ME1654-1	Innovation and Design Thinking

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Part I**LINEAR ALGEBRA AND NUMERICAL METHODS****1 MSE-I Examinations****1.1 MA2005-1 MSE-I – February 2025****Mid Semester Examination - I, February 2025**

Duration: 1 Hour — Max Marks: 15

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS**Part A: Multiple Choice Questions ($3 \times 1 = 3$ Marks)**

- For what values of k will the augmented matrix $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 2 & 1 & 3 \\ 1 & 2 & k & k+1 \end{array} \right]$ correspond to an inconsistent system of linear equations?
(A) $k \neq 1$ (B) $k = 0$ (C) $k = 1$ (D) All values of k
- If $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, find $(AB)^{-1}$.
(A) $\begin{bmatrix} 1 & 1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1/2 \\ -1/6 & 1/2 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & -1/3 \\ 1/6 & 1/2 \end{bmatrix}$
- Given a matrix A , column 1 and column 3 are pivot columns, then the dimension of $\text{Col } A$ is:
(A) 1 (B) 2 (C) 3 (D) 4

Part B: Descriptive Questions ($2 \times 6 = 12$ Marks)**UNIT – I**

Q1. a) Find the eigenvalues of $A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$. Also determine the eigenvectors corresponding to each eigenvalue. [6 Marks]

b) Reduce the matrix $A = \begin{bmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$ to the form $PA = LU$, using partial pivoting, where L is a lower triangular matrix with unit diagonal entries and U is an upper triangular matrix. [6 Marks]

UNIT – II

Q2. a) Solve by Gauss-Seidel method: $10x + y + z = 12$; $2x + 10y + z = 13$; $x + y + 5z = 7$. (Perform 3 iterations) [6 Marks]

b) Find the dominant eigenvalue and corresponding eigenvector of $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ using Rayleigh's Power method (4 iterations). [6 Marks]

1.2 MA2005-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024

Duration: 1 Hour — Max Marks: 15

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part B: Descriptive Questions

Q1. a) Solve by Gauss-Seidel iteration method (4 iterations): $20x + y - 2z = 17$; $3x + 20y - z = -18$; $2x - 3y + 20z = 25$. Start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$.

b) Using the power method, find the dominant eigenvalue and corresponding eigenvector of $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}$ starting with initial value $[1, 0, 0]^T$. (4 iterations)

Q2. a) Solve the system $x + y + z = 1$; $4x + 3y - z = 6$; $3x + 5y + 3z = 4$ by LU factorization method.

b) Find all eigenvalues of $A = \begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}$. Find eigenvector for largest eigenvalue.

Q3. a) Find the general solution of the system whose augmented matrix is $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$.

b) Reduce the matrix $A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$ to echelon form and find its rank.

1.3 21CS401 MSE-I – June 2023

Mid Semester Examination - I, June 2023

Duration: 1 Hour — Max Marks: 20

21CS401 – LINEAR ALGEBRA

Part B: Descriptive Questions

Q1. a) Find the eigenvalues and eigenvectors of the matrix.

b) Reduce the given matrix to echelon form.

Q2. a) Solve the system of equations using Gauss elimination method.

b) Solve using LU factorization method.

Q3. a) Apply Power method to find dominant eigenvalue (4 iterations).

b) Diagonalize the matrix $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$. Hence find A^6 .

2 MSE-II Examinations

2.1 MA2005-1 MSE-II – April 2025

Mid Semester Examination - II, April 2025

Duration: 1 Hour — Max Marks: 20

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

- Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, x_1)$. Then $T^{-1}(x_1, x_2)$ equals:
(A) $(x_2, x_1 - x_2)$ (B) $(x_2, x_1 + x_2)$ (C) $(x_1, x_1 - x_2)$ (D) $(x_1, x_1 + x_2)$
- The set $\{1, x, x^2\}$ forms a basis for:
(A) P_1 (B) P_2 (C) P_3 (D) P_4
- Which of the following is a subspace of the given vector space?
(A) The set of all polynomials of degree exactly 3, in P_3
(B) The set of all polynomials with positive coefficients, in P
(C) The set of all 2×2 matrices with positive entries, in $M_{2 \times 2}$
(D) The set of all symmetric 2×2 matrices, in $M_{2 \times 2}$
- Given matrix $A_{4 \times 5}$ with $\text{rank}(A) = 3$, then $\dim(\text{Null}(A))$ is:
(A) 1 (B) 2 (C) 3 (D) 4

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)**UNIT – I**

Q1. a) Let $T : P_2 \rightarrow P_2$ be the transformation that maps a polynomial $p(t)$ into the polynomial $p(t+2)$. Find the image of $p(t) = 3 - 2t + t^2$. Show that T is a linear transformation. Find the matrix representation of T relative to the basis $B = \{1, t, t^2\}$. [8 Marks]

b) Let $\vec{v}_1 = (1, -2, 3)$, $\vec{v}_2 = (5, 6, -1)$, $\vec{v}_3 = (3, 2, 1)$. Show that the set $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis for $\mathbb{R}^3$. Express each of the standard basis vectors as linear combinations of the vectors in S . [8 Marks]

UNIT – II

Q2. a) Use Gram-Schmidt process to construct orthonormal basis from the basis $B = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ of $\mathbb{R}^3$. Find the QR factorization of the matrix whose column vectors form B . [8 Marks]

b) Use coordinate vectors to test the linear independence of the set of polynomials: $S = \{1 - 2t^2 - t^3, t + 2t^3, 1 + t - 2t^2\}$. [8 Marks]

2.2 21CS401 MSE-II – July 2023
Mid Semester Examination - II, July 2023
21CS401 – LINEAR ALGEBRA

Duration: 1 Hour — Max Marks: 20

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

- The dimension of the vector space of all symmetric $n \times n$ matrices is:
(A) n^2 (B) $\frac{n(n+1)}{2}$ (C) $\frac{n(n-1)}{2}$ (D) n
- Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (x + y, x - y)$. Then T is:
(A) Linear but not one-to-one (B) One-to-one but not onto
(C) Both one-to-one and onto (D) Neither one-to-one nor onto

3. If $\{v_1, v_2, v_3\}$ is a basis for V , then $\{v_1 + v_2, v_2 + v_3, v_3 + v_1\}$ is:
 (A) Always a basis (B) Never a basis (C) A basis if vectors are orthogonal (D) Cannot be determined
4. The null space of an $m \times n$ matrix A is a subspace of:
 (A) $\mathbb{R}^m$ (B) $\mathbb{R}^n$ (C) $\mathbb{R}^{m+n}$ (D) $\mathbb{R}^{mn}$

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)

Q1. a) Let $V = P_2$ (polynomials of degree ≤ 2). Show that $S = \{1 + x, x + x^2, 1 + x^2\}$ is a basis for V . [8 Marks]

b) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be defined by $T(x, y, z) = (x + 2y - z, 2x - y + z)$. Find the matrix of T with respect to standard bases. Find $\ker(T)$ and $\text{range}(T)$. [8 Marks]

Q2. a) Use Gram-Schmidt process to orthonormalize the vectors $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$. [8 Marks]

b) Find the least squares solution of $A\vec{x} = \vec{b}$ where $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$. [8 Marks]

2.3 20CS401/20IS401/20AM401/20CC401 MSE-II – June 2022

Mid Semester Examination - II, June 2022

Duration: 1 Hour — Max Marks: 20

20CS401/20IS401/20AM401/20CC401 – LINEAR ALGEBRA & PROBABILITY THEORY

Note: Answer any One full question from each Unit.

UNIT – I

Q1. a) Check whether a transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (3x + y, 5x + 7y, x + 3y)$ is a linear transformation or not. [4 Marks]

b) State Rank-Nullity theorem. [4 Marks]

Q2. a) Let a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Find the null space of T and rank of T . [6 Marks]

Q3. Find the matrix of the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x - y + 2z, 3x + y)$ with respect to bases $B = \{(1, 1, 1), (1, 2, 3), (1, 0, 0)\}$ and $B' = \{(1, 1), (1, -1)\}$. [4 Marks]

Q4. Construct an orthonormal basis for a subspace W of $\mathbb{R}^3$ with a basis $\{x_1, x_2, x_3\}$ where $x_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $x_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $x_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ of W , using Gram-Schmidt process. [6 Marks]

UNIT – II

Q5. Find a least square solution of $AX = B$ where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$. [4 Marks]

Q6. a) The pdf $p(x)$ of a variate X is given by the following table:

X	0	1	3	4	5	6
$p(x)$	k	$3k$	$5k$	$9k$	$10k$	$12k$

For what value of k , does this represent a valid probability distribution? Also find $P(X > 3)$ and $E[X]$. [3+3 Marks]

b) If $f(x) = kx^3$, $0 < x < 1$, and $= 0$ elsewhere, is a pdf of the random variable X , find:

- (i) the value of k
- (ii) $P(X > 0.7)$
- (iii) $P\left(\frac{1/4 < X < 3/4}{X > 1/2}\right)$

[3+3 Marks]

Q7. a) If X is a random variable, prove that:

- (i) $V(X) = E(X^2) - [E(X)]^2$
- (ii) $V(aX + b) = a^2V(X)$ where a and b are constants.

[4 Marks]

Q8. a) State Bayes' theorem. [2 Marks]

b) A business man goes to hotels X , Y , Z 30%, 45% and 25% of the time respectively. It is known that 5%, 4%, 6% of the rooms in hotels X , Y , Z respectively have faulty plumbing. Determine the probability:

- (i) that the business man goes to hotel with faulty plumbing
- (ii) that business man's room having faulty plumbing is assigned to hotel Y ?

[6 Marks]

3 SEE (Semester End Examinations)

3.1 MA2005-1 SEE – May 2025

Semester End Examination, May 2025

Duration: 3 Hours — Max Marks: 100

MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS

Part A: Multiple Choice Questions ($20 \times 1 = 20$ Marks)

- For what value of k , the augmented matrix $\left[\begin{array}{ccc|c} 1 & 3 & -2 & 3 \\ 3 & k & -6 & 9 \\ 1 & -3 & 2 & 5 \end{array} \right]$ corresponds to an inconsistent system?
 (A) $k = 1$ (B) $k = 9$ (C) $k = -3$ (D) None of these
- If A and B are two matrices such that AB exists and is a square matrix, then:
 (A) A and B are square matrices of same order
 (B) Either both are square or number of rows of A = number of columns of B
 (C) BA need not exist
 (D) Any one of A and B is identity
- Columns 1, 2 and 5 are pivot columns of a matrix A . Which is NOT necessarily true?
 (A) $\dim(\text{Col } A) = 3$ (B) $\dim(\text{Row } A) = 3$ (C) $\text{rank}(A) = 3$ (D) $\dim(\text{Null } A) = 3$
- If λ is an eigenvalue of A , then eigenvalue of A^{-1} is:
 (A) λ (B) $-\lambda$ (C) λ^2 (D) $1/\lambda$

5. If $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ then $A^3 - 3A^2 - A + 3I =$
 (A) $A + I$ (B) I (C) $2A$ (D) 0
6. The eigenvalues of $P^{-1}AP$ are:
 (A) Same as that of PA (B) Same as that of A (C) Same as that of P (D) Cannot say
7. Correct order to solve by Gauss-Seidel method:
 (A) Arrange in any order (B) Diagonal elements are zero
 (C) Diagonal elements larger than other elements (D) Cannot be solved
8. Which is a direct method for solving system of linear equations?
 (A) Jacobi (B) Gauss-Seidel (C) LU Decomposition (D) Power method
9. If λ is eigenvalue of A with eigenvector $\vec{x}$, then $\lambda^2 + 1$ is eigenvalue of:
 (A) $A^2 + I$ (B) $A^2 + A$ (C) $2A$ (D) $(A + I)^2$
10. If $1 + x$, $1 + x + x^2$, and $ax^2 + x^3$ are linearly dependent, then $a =$:
 (A) 1 (B) 2 (C) 3 (D) 4
11. The dimension of subspace $W = \{(a, b, c, d) : a - 2b + c - d = 0\}$ is:
 (A) 1 (B) 2 (C) 3 (D) 4
12. Let V be set of all 2×2 symmetric matrices. Then dimension of V is:
 (A) 1 (B) 2 (C) 3 (D) 4
13. For linear transformation $T : V \rightarrow W$, let U be subspace of W . Then $T^{-1}(U)$ is:
 (A) Subspace of V (B) Subspace of W (C) Subspace of both (D) Not a subspace
14. Every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be represented as matrix multiplication.
 (A) True only if $m = n$ (B) True only if $m \neq n$ (C) Always true (D) Never true
15. If $\vec{u}$ and $\vec{v}$ are orthogonal vectors in $\mathbb{R}^n$, then:
 (A) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$ (B) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 - \|\vec{v}\|^2$
 (C) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 \cdot \|\vec{v}\|^2$ (D) $\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 / \|\vec{v}\|^2$
16. Two vectors $\vec{u} = (1, 2, -3)$ and $\vec{v} = (a, 1, 1)$ are orthogonal if $a =$:
 (A) 1 (B) 2 (C) 3 (D) 5
17. A linear combination of the rows of A can be written as:
 (A) $\vec{x}^T A$ (B) $A \vec{x}$ (C) $\vec{x} A^T$ (D) $A^T \vec{x}$
18. The angle between vectors $\vec{u} = (1, 2, -1, 0)$ and $\vec{v} = (-1, 1, 1, 2)$ is:
 (A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) $2\pi/3$
19. Which defines an inner product on $\mathbb{R}^2$?
 (A) $\langle \vec{u}, \vec{v} \rangle = u_1 v_1 - u_2 v_2$ (B) $\langle \vec{u}, \vec{v} \rangle = u_1 v_1 + 2u_2 v_2$
 (C) $\langle \vec{u}, \vec{v} \rangle = u_1^2 v_1^2 + u_2^2 v_2^2$ (D) $\langle \vec{u}, \vec{v} \rangle = u_1 v_2 - u_2 v_1$
20. For orthonormal basis $\{\vec{u}_1, \dots, \vec{u}_n\}$ of $\mathbb{R}^n$ and any $\vec{x} \in \mathbb{R}^n$:
 (A) $\|\vec{x}\|^2 = \sum_{i=1}^n |\langle \vec{x}, \vec{u}_i \rangle|^2$ (B) $\|\vec{x}\|^2 = \sum_{i=1}^n \langle \vec{x}, \vec{u}_i \rangle$
 (C) $\|\vec{x}\| = \sum_{i=1}^n |\langle \vec{x}, \vec{u}_i \rangle|$ (D) None of the above

Part B: Descriptive Questions ($5 \times 16 = 80$ Marks)**UNIT – I**

Q1. a) Find basis for Nul A and Col A where $A = \begin{bmatrix} 1 & 3 & 0 & 2 & -1 \\ 0 & 0 & 1 & -1 & 3 \\ 1 & 3 & 1 & 1 & 2 \\ -2 & -6 & 0 & -2 & 4 \end{bmatrix}$ [8 Marks]

b) Find eigenvalues of $A = \begin{bmatrix} 0 & -4 & -6 \\ -1 & 0 & -3 \\ 1 & 2 & 5 \end{bmatrix}$. Find eigenvectors and verify Cayley-Hamilton theorem. [8 Marks]

Q2. a) Compute A^{10} if $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$. [8 Marks]

b) Reduce $A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$ to $A = LU$. Use factorization to solve $A\vec{x} = \vec{b}$ where $\vec{b} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$. [8 Marks]

UNIT – II

Q3. a) Solve by Gauss-Seidel method (4 iterations): $10x - y + 2z = 6$; $-x + 11y - z + 3w = 25$; $2x - y + 10z - w = -11$; $3y - z + 8w = 15$. [8 Marks]

b) Find dominant eigenvalue and eigenvector of $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using Power method (5 iterations). [8 Marks]

Q4. a) Let $V = M_{2 \times 2}$. Show $W = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a + d = 0 \right\}$ is subspace of V . Find its dimension. [8 Marks]

b) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be $T(x_1, x_2) = (x_1 - x_2, x_1, 2x_1 + x_2)$. Show T is linear. Find matrix. Is T one-to-one? Is T onto? [8 Marks]

UNIT – III

Q5. a) Find QR factorization of $A = \begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$. [8 Marks]

b) For $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$, find the SVD of A . [8 Marks]

Q6. a) Find best line $y = \beta_0 + \beta_1 x$ for data points: $(0, 1), (1, 1), (2, 2), (3, 2)$ using least squares. [8 Marks]

b) Let $S = \{(2, 5, -3, -1), (-2, -3, 2, -5), (1, 3, -2, 0), (-1, -6, 3, 3)\}$. Find basis for subspace of $\mathbb{R}^4$ spanned by S . [8 Marks]

Part II**DATABASE MANAGEMENT SYSTEMS****4 MSE-I Examinations****4.1 CS2102-1 MSE-I – February 2025**

Mid Semester Examination - I, February 2025
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Duration: 1 Hour — Max Marks: 15

Part A: Multiple Choice Questions ($3 \times 1 = 3$ Marks)

1. A _____ is a unit of storage containing one or more records.
 (A) Block (B) Field (C) Tuple (D) File
2. Which of the following is NOT a component of ER Model?
 (A) Entity (B) Attribute (C) Table (D) Relationship
3. Which of the following is used to uniquely identify a record in a table?
 (A) Foreign Key (B) Candidate Key (C) Primary Key (D) Super Key

Part B: Descriptive Questions ($2 \times 6 = 12$ Marks)**UNIT – I**

Q1. a) Design and construct an ER diagram for “Student Database Management System”. Indicate all keys, constraints and assumptions. [6 Marks]

b) Discuss the main characteristics of the database approach. [6 Marks]

Q2. a) Identify the Entity, Relationship, Participation Constraint and Cardinality ratio for the ER diagram of Movie database schema (ACTOR, MOVIE, DIRECTOR, PRODUCER entities with relationships). [6 Marks]

b) Describe Three Schema Architecture with a neat diagram. [6 Marks]

UNIT – II

Q3. a) Write the following queries in Relational Algebra for:

EMPLOYEE(SSN, SNAME, AGE, SALARY, DEPT_ID)

DEPARTMENT(DEPT_ID, DEPT_NAME, DEPT_HEAD)

PROJECT(PID, DEPT_ID, PNAME)

WORKSON(SSN, PID, PHOURS)

- (i) Get names of employees working on projects managed by 'HR' department.
 - (ii) Retrieve names of employees who do not work on any project.
 - (iii) Find names of all departments that have at least one project. [6 Marks]
- b) Explain how to deal with constraints violation during insertion and delete operations. [6 Marks]
-

4.2 CS2102-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024
CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Duration: 1 Hour — Max Marks: 15

Part B: Descriptive Questions

- Q1.** a) Describe Three Schema Architecture with diagram. Why do we need mappings among schema levels? [6 Marks]
 b) Design ER diagram for “Engineering College Management System”. [6 Marks]
Q2. a) Discuss the main characteristics of database approach. [6 Marks]
 b) Explain domain constraints and referential integrity constraints in relational model. [6 Marks]
Q3. a) Define Entity attribute. Explain different types of attributes in ER diagram. [6 Marks]
 b) Outline steps to convert basic ER model to relational database scheme. [6 Marks]
Q4. a) “For good database design, matching attributes must be primary key or foreign key otherwise it leads to spurious tuples”. Justify. [6 Marks]
 b) Discuss the characteristics of relations. [6 Marks]

5 MSE-II Examinations**5.1 CS2102-1 MSE-II – April 2025****Mid Semester Examination - II, April 2025**

Duration: 1 Hour — Max Marks: 20

CS2102-1 – DATABASE MANAGEMENT SYSTEMS**Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)**

- The rule “No component of Primary Key accepts Null value” is:
 (A) Referential Integrity Constraint (B) Domain Integrity Constraint
 (C) Entity Integrity Constraint (D) Tuple Integrity Constraint
- Which command adds a new column to existing table?
 (A) ALTER (B) UPDATE (C) INSERT (D) MODIFY
- A relation is in BCNF if every non-trivial FD $X \rightarrow Y$:
 (A) X is a super key (B) Y is a prime attribute
 (C) X is a candidate key (D) Y is a super key
- The operation to combine tuples from two relations based on condition:
 (A) Union (B) Join (C) Projection (D) Selection

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)**Q1.** Consider the schema and write SQL queries:

Suppliers(sid: integer, sname: string, address: string)

Parts(pid: integer, pname: string, color: string)

Catalog(sid: integer, pid: integer, cost: double)

- Find the sids of suppliers who supply some red or green part.
 - Find the sids of suppliers who supply only red parts.
 - Find the pnames of parts for which there is some supplier.
 - Find the sids of suppliers who supply some red part or are at 221 Packer Street. (Nested Query)
- [8 Marks]

Q2. Consider the schema and write SQL queries:

```

Movie(movie_id, title, release_year, genre, director_id, rating)
Director(director_id, name, birth_year, nationality)
Actor(actor_id, name, birth_year, nationality)
Movie_Actor(movie_id, actor_id, role)
Review(rev_id, user_id, movie_id, rev_text, rating, rev_date)
User(user_id, username, email, age)

```

- (i) Find names of actors who acted in at least one movie released after 2015.
 - (ii) Find directors who directed more than 5 movies.
 - (iii) Get average rating of movies directed by 'Christopher Nolan'.
 - (iv) Find movies with higher rating than average rating of same genre. (Nested Query) [8 Marks]
- Q3.** a) Consider FDs $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$. Are they equivalent? [8 Marks]
- b) What is Normalization? Explain First Normal Form with example. [8 Marks]
- Q4.** a) Given $R(ABCDEFGHIJ)$ and FDs: $\{(A, B) \rightarrow C, A \rightarrow \{D, E\}, B \rightarrow F, F \rightarrow \{G, H\}, D \rightarrow \{I, J\}\}$. Find candidate key. Check if relation is in 3NF. If not, decompose to 3NF. [8 Marks]
- b) Explain any two Informal Design Guidelines for Relation Schemas. [8 Marks]

6 SEE (Semester End Examinations)

6.1 CS2102-1 SEE – May 2025

Semester End Examination, May 2025

Duration: 3 Hours — Max Marks: 100

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part A: Multiple Choice Questions ($20 \times 1 = 20$ Marks)

1. Which is not a characteristic of DBMS? (A) Data Independence (B) Concurrent Access (C) Data Redundancy (D) Data Security
2. Weak entity is represented by: (A) Rectangle (B) Double Rectangle (C) Diamond (D) Oval
3. Grouping related objects in ER model: (A) Aggregation (B) Generalization (C) Specialization (D) Composition
4. Each row of table represents a: (A) Field (B) Record (C) Attribute (D) Key
5. Retrieve data from multiple tables: (A) Union (B) Intersection (C) Cartesian Product (D) Projection
6. Produces relation with only specified attributes: (A) Selection (B) Projection (C) Join (D) Division
7. Filter groups formed by GROUP BY: (A) WHERE (B) ORDER BY (C) HAVING (D) FILTER
8. NOT a DML command: (A) SELECT (B) INSERT (C) CREATE (D) UPDATE
9. Trivial FD $X \rightarrow Y$ if: (A) $X \subseteq Y$ (B) $Y \subseteq X$ (C) $X = Y$ (D) $X \cap Y = \phi$
10. Eliminates transitive dependencies: (A) 1NF (B) 2NF (C) 3NF (D) BCNF
11. Closure of attributes X under FDs F : (A) X^+ (B) F^+ (C) X^* (D) F^*
12. 2NF requires 1NF and: (A) No partial dependencies (B) No transitive dependencies (C) Every determinant is candidate key (D) All attributes atomic

13. Lossless join decomposition uses: (A) Union (B) Intersection (C) Natural join (D) Cartesian product
14. NOT property of good decomposition: (A) Lossless join (B) Dependency preservation (C) Redundancy (D) No anomalies
15. Minimal cover has: (A) Single attribute on RHS (B) No redundant FDs (C) No extraneous attributes on LHS (D) All of the above
16. Permanently save transaction changes: (A) SAVEPOINT (B) COMMIT (C) ROLLBACK (D) GRANT
17. ACID properties include: (A) Atomicity, Consistency, Isolation, Durability (B) Atomicity, Concurrency, Isolation, Durability (C) Accuracy, Consistency, Integrity, Durability (D) Atomicity, Consistency, Integrity, Durability
18. Conflict serializable if: (A) Can transform to serial schedule by swapping non-conflicting operations (B) All transactions execute serially (C) No conflicts (D) Precedence graph has no cycles
19. Guarantees deadlock freedom: (A) Simple Lock (B) Two-Phase Locking (C) Strict 2PL (D) Wait-Die Scheme
20. Timestamp ordering for write on X : (A) Always allowed (B) If $TS > W\text{-timestamp}(X)$ and $TS > R\text{-timestamp}(X)$ (C) If $TS < W\text{-timestamp}(X)$ (D) Never allowed

Part B: Descriptive Questions (80 Marks)

- Q1.** a) What are Characteristics of Database approach? Explain. [8 Marks]
 b) With neat diagram, explain Three Schema architecture. [8 Marks]
- Q2.** a) Explain attribute types: (i) Composite (ii) Multivalued (iii) Atomic. [8 Marks]
 b) Explain Cardinality Ratio for Binary relationships with examples. [8 Marks]
- Q3.** a) Consider STUDENT and COURSE tables. Write Relational Algebra queries. [8 Marks]
 b) Explain steps in ER to Relational Mapping Algorithm. [8 Marks]
- Q4.** a) Explain Not Null, DEFAULT and CHECK constraints in SQL. [8 Marks]
 b) Explain CASCADE and RESTRICT options of DROP TABLE. [8 Marks]
- Q5.** a) What are Informal Design Guidelines for relational schema? [8 Marks]
 b) What is Functional Dependency? Explain with example. [8 Marks]
- Q6.** a) Explain Standard Inference rules for Functional Dependencies. [8 Marks]
 b) Explain algorithm for determining X^+ , Closure of X under F . [8 Marks]
- Q7.** a) What is B+ Tree? Explain structure & characteristics with diagram. [8 Marks]
 b) Briefly explain ACID properties. [8 Marks]
- Q8.** a) What is Query Optimizer? Explain working with diagram. [8 Marks]
 b) What are 2 rules of Strict 2PL? Explain. [8 Marks]

6.2 CS2102-1 SEE – July 2024

Semester End Examination, July 2024

Duration: 3 Hours — Max Marks: 100

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

Part B: Descriptive Questions

- Q1.** a) Briefly explain various characteristics of database approach. [8 Marks]
b) Explain Relational Algebra operations: (i) Division (ii) Union (iii) Cartesian product (iv) COUNT. [8 Marks]
- Q2.** a) Design ER diagram and schema for flight management system. [8 Marks]
b) Explain grouping concepts in relational algebra. [8 Marks]
- Q3.** a) With block diagram explain three schema architecture. [8 Marks]
b) Sailors-Boats-Reserves queries in Relational Algebra. [8 Marks]
- Q4.** a) Employee-Works-Company-Manages SQL queries. [8 Marks]
b) What is a view? Give specification in SQL. [8 Marks]
- Q5.** a) Check lossless join property for given decompositions. [8 Marks]
b) How basic constraints are specified in SQL-99. [8 Marks]
- Q6.** a) Order processing database SQL queries. [8 Marks]
b) Need for normalization. Explain 1NF, 2NF, 3NF with examples. [8 Marks]
- Q7.** a) Explain Strict Two-Phase Locking protocol. [8 Marks]
b) Explain index structures: Hash based, Tree based indexing. [8 Marks]
- Q8.** a) Illustrate anomalies with Interleaved Execution. [8 Marks]
b) Algorithm to insert element in B+ tree. [8 Marks]

Part III

DESIGN AND ANALYSIS OF ALGORITHMS

7 MSE-I Examinations

7.1 CS3004-1 MSE-I – February 2025

Mid Semester Examination - I, February 2025

Duration: 1 Hour — Max Marks: 15

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part A: Multiple Choice Questions ($3 \times 1 = 3$ Marks)

1. If algorithm has $T(n) = 7n^3 + 10n^2 + 4$, then $T(n)$ belongs to:
(A) $O(n)$ (B) $O(n^2)$ (C) $O(n^3)$ (D) $O(n^4)$
2. Time complexity of $T(n) = 2T(n/2) + n$:
(A) $O(n)$ (B) $O(n \log n)$ (C) $O(n^2)$ (D) $O(\log n)$
3. Minimum comparisons to find minimum in array of n elements:
(A) n (B) $n - 1$ (C) $n + 1$ (D) $2n$

Part B: Descriptive Questions ($2 \times 6 = 12$ Marks)

UNIT – I

- Q1.** a) Summarize algorithm design and analysis process with diagram. [6 Marks]
 b) For given algorithm, determine: (i) What it computes (ii) Basic operation (iii) How many times executed (iv) Efficiency. [6 Marks]

Q2. a) Define asymptotic notations Big Oh (O) and Big Theta (Θ) with examples. [6 Marks]

b) Show that efficiency of Tower of Hanoi is exponential. [6 Marks]

UNIT – II

Q3. a) Develop brute force string matching algorithm. Analyze worst-case complexity. [6 Marks]

b) Outline merge sort algorithm and derive time complexity. [6 Marks]

Q4. a) Define TSP. Solve using exhaustive search for 4-vertex graph with edges: A-B=6, A-C=3, A-D=1, B-C=4, B-D=2, C-D=8. State efficiency. [6 Marks]

b) Apply Quick sort on marks: 8, 10, 5, 2, 14. Show tree of recursive calls. [6 Marks]

7.2 CS3004-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024

Duration: 1 Hour — Max Marks: 15

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part B: Descriptive Questions

- Q1.** a) Explain properties of algorithm using linear search example. [6 Marks]
 b) For each function, indicate change when argument increased fourfold: (i) $\log_2 n$ (ii) n (iii) n^2 (iv) 2^n (v) $\sqrt{n}$. [6 Marks]

- Q2.** a) Explain general framework for Analysis of an algorithm. [6 Marks]
 b) Define 3 asymptotic notations. Check: (i) $10n^2 + 9 \in O(n^2)$ (ii) $n(n+1)/2 \in O(n^2)$. [6 Marks]
Q3. a) Write general plan for analyzing recursive algorithm. Find factorial efficiency. [6 Marks]
 b) Give two brute force algorithms to sort integers. [6 Marks]
Q4. a) Illustrate Merge sort and analyze efficiency. Sort: 6, 14, 8, 2, 15, 10, 7, 5. [6 Marks]
 b) Show comparisons for brute force string matching: P=0001 in T=000010001010101. [6 Marks]

7.3 21CS402 MSE-I – June 2023

Mid Semester Examination - I, June 2023

Duration: 1 Hour — Max Marks: 20

21CS402 – DESIGN AND ANALYSIS OF ALGORITHMS

Part B: Descriptive Questions

- Q1.** a) With process flow diagram explain algorithm design and analysis. [8 Marks]
 b) Design recursive algorithm for $2^n = 2^{n-1} + 2^{n-1}$. Find complexity. [8 Marks]
Q2. a) Define three asymptotic notations. Express $f(n) = n(n-1)/2$ using three notations with proof. [8 Marks]
 b) Perform complexity analysis of sequential search for three efficiency types. [8 Marks]
Q3. a) What is brute force? Give brute force string matching with complexity. Trace: TEXT: JIM SAW ME IN SHOE SHOP, PATTERN: SHE. [8 Marks]
 b) Apply merge sort to L,A,S,T,S,O,R,T,I,N,G alphabetically. Analyze complexity. [8 Marks]
Q4. a) Give complete Quick sort algorithm with complexity analysis. Trace for: 5, 3, 1, 9, 8, 2, 4, 7. [8 Marks]
 b) What is exhaustive search? Solve knapsack: $n = 4$, capacity $m = 10$, $w = (4, 6, 3, 5)$, values $= (30, 50, 20, 10)$. [8 Marks]

8 MSE-II Examinations

8.1 CS3004-1 MSE-II – April 2025

Mid Semester Examination - II, April 2025

Duration: 1 Hour — Max Marks: 20

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

- Necessary condition for topological sorting:
 (A) Graph must be tree (B) Graph must be DAG
 (C) Graph must be complete (D) Graph must be bipartite
- Data structure used in DFS:
 (A) Queue (B) Heap (C) Priority queue (D) Stack
- Time complexity of distribution counting sort:
 (A) $\Theta(n^2)$ (B) $O(n \log n)$ (C) $O(2^n)$ (D) $\Theta(n)$

4. Dynamic programming builds solution from:
- (A) Overlapping subproblems (B) Single recursive call
(C) Greedy approach (D) Divide and conquer

Part B: Descriptive Questions ($2 \times 8 = 16$ Marks)

- Q1.** a) Outline Insertion sort algorithm. Show sorting of E, X, A, M, P, L, E in alphabetical order. [4 Marks]
b) Apply heap sort to: 90, 89, 56, 72, 67, 12, 21, 34, 45. [4 Marks]
- Q2.** a) Define AVL tree. Construct AVL tree for {2, 9, 3, 12, 15, 8, 11}. [4 Marks]
b) Define topological sorting. Illustrate using Source Removal method for given graph. [4 Marks]
- Q3.** a) Write Comparison Counting sort algorithm. Compare complexity with Selection and Bubble sort. [4 Marks]
b) Summarize Horspool algorithm. Search TIME in “A STITCH IN TIME SAVES NINE”. [4 Marks]
- Q4.** a) Apply All Pair Shortest Path algorithm for given weighted graph. [4 Marks]
b) Write BFS algorithm. List two facts differentiating BFS from DFS. [4 Marks]

8.2 20CS402/20IS402/20AM402/20CC402 MSE-II – June 2022

Mid Semester Examination - II, June 2022

Duration: 1 Hour — Max Marks: 20

20CS402/20IS402/20AM402/20CC402 – DESIGN AND ANALYSIS OF ALGORITHMS

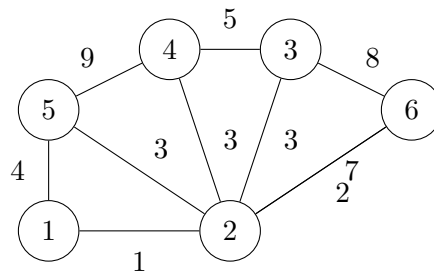
Note: Answer any One full question from each Unit.

UNIT – I

- Q1.** Develop an algorithm to perform Insertion Sort and analyze the best and worst case efficiencies. [4 Marks]
- Q2.** Find the solution for following problem using topological sorting method. Consider a set of 5 courses $\{C_1, C_2, C_3, C_4, C_5\}$, a part time student has to take in some degree program. The courses can be taken in any order as long as the following course prerequisites are met: C_1 and C_2 have no prerequisites, C_3 requires C_1 and C_2 , C_4 requires C_3 and C_5 requires C_3 and C_4 . The student can take only one course per term. [3 Marks]
- Q3.** Construct 2-3 tree for the data: C, O, M, P, U, T, I, N, G. [3 Marks]
- Q4.** Explain Horspool’s algorithm and apply the algorithm to find the pattern “MONEY” in the text “A FOOL AND HIS MONEY ARE PARTED SOON”. [4 Marks]
- Q5.** Apply closed hashing to distribute the keys 10, 23, 14, 19, 17, 32, 45 using the function $H(k) = K \bmod 11$. [2 Marks]
- Q6.** Construct heap and sort using array representation: 2, 9, 8, 14, 22, 16, 7. [4 Marks]

UNIT – II

- Q7.** Outline the pseudocode for Prim’s algorithm to find the minimum spanning tree. Apply the Prim’s algorithm to find the minimum spanning tree for the following graph:



[6 Marks]

Q8. Write a pseudocode for Bellman Ford algorithm to find the single source shortest path. [4 Marks]

Q9. a) Construct a Huffman code for the following data:

Character	A	B	C	D	-
Probability	0.4	0.1	0.2	0.15	0.15

Encode the text “ABACABAD” using the code obtained. Decode the text whose encoding is 1000101110010010. [6 Marks]

b) Write the pseudocode for single source shortest path in DAG. [4 Marks]

9 SEE (Semester End Examinations)

9.1 CS3004-1 SEE – May 2025

Semester End Examination, May 2025

Duration: 3 Hours — Max Marks: 100

CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS

Part A: Multiple Choice Questions ($20 \times 1 = 20$ Marks)

- Quick sort divides array into: (A) One (B) Two (C) Three (D) Four
- Binary Search works on: (A) Unsorted (B) Sorted (C) Distinct elements (D) Integers only
- Linear search is also called: (A) Binary search (B) Simple search (C) Sequential search (D) Difficult search
- $O(n)$ means: (A) Exponential (B) Linear growth (C) Independent of input (D) Logarithmic
- Output of fun(5) where fun(n) returns 1 if $n==1$, else $n+fun(n-1)$: (A) 120 (B) 24 (C) 6 (D) 15
- Pseudocode uses: (A) Native + programming language (B) High level only (C) Natural language only (D) Low level
- NOT a characteristic of good algorithm: (A) Clarity (B) Efficiency (C) Correctness (D) Complex data structures
- Primary characteristic of Divide and Conquer: (A) Iteration (B) Overlapping subproblems (C) Dividing into independent subproblems, solving recursively, combining (D) Linear solving
- 0/1 Knapsack uses: (A) Greedy (B) Divide and conquer (C) Dynamic programming (D) DFS
- Floyd-Warshall finds: (A) Shortest path between two nodes (B) Shortest paths between all pairs (C) Longest path (D) Pattern matching

11. Horspool uses which table: (A) Shift Table (B) Hash Table (C) Match Table (D) Character Table
12. Counting Sort is distribution-based because: (A) Uses frequencies for positions (B) Divides into subarrays (C) Compares pairs (D) Recursively splits
13. Max-heap root from [3, 9, 2, 1, 4, 8]: (A) 1 (B) 9 (C) 2 (D) 8
14. AVL rotation when balance factor > 1 or < -1 : (A) Rotation (B) Deletion (C) Insertion (D) Search
15. Binary Search is: (A) Decrease by constant (B) Decrease by constant factor (C) Decrease by one (D) Divide and conquer
16. In max-heap: (A) Parent smaller (B) Parent greater or equal (C) BST (D) Ascending order
17. Job Assignment minimizes: (A) Time (B) Number of tasks (C) Total cost (D) Total time worked
18. Subset-sum backtracks when: (A) Sum exceeds target (B) Sum equals target (C) All elements considered (D) Subset found
19. Dijkstra's Algorithm is: (A) Greedy (B) Branch and bound (C) Backtracking (D) Dynamic programming
20. Huffman codes use: (A) Exhaustive search (B) Greedy (C) Brute force (D) Divide and conquer

Part B: Descriptive Questions (80 Marks)

UNIT – I

Q1. a) Explain Performance Analysis of algorithm. [8 Marks]

b) Write Selection sort pseudocode and analyze. Sort: 5, 4, 2, 1, 6, 3. [8 Marks]

Q2. a) General plan for non-recursive algorithm analysis. Find maximal element and analyze. [8 Marks]

b) Write Merge sort algorithm. Derive time complexity using recurrence. Sort: 38, 27, 43, 3, 9, 82, 10. [8 Marks]

UNIT – II

Q3. a) Write insertion sort. Construct heap for: 63, 9, 19, 7, 90, 81, 45. [8 Marks]

b) Identify "JIM_SAM" using Horspool in "JIM_SAW_ME_IN_A_BARBERSHOP". [8 Marks]

Q4. a) Write B-Tree insertion algorithm. Draw B-Tree of order 5 for: 3, 14, 7, 1, 8, 5, 11, 17, 13, 6, 23, 12, 20, 26, 4, 16, 18, 24, 25, 19. [8 Marks]

b) Define Topological Sorting. Illustrate using source removal method. [8 Marks]

UNIT – III

Q5. a) Write Divide and Conquer algorithm for key comparisons. Derive complexity. Quick sort: 38, 27, 43, 3, 9, 82, 10. [8 Marks]

b) Define Class P, Class NP, NP-Complete, NP-Hard. [8 Marks]

Q6. a) Solve Job assignment using Branch and Bound for matrix: [8 Marks]

$$\begin{bmatrix} 9 & 2 & 7 & 8 \\ 6 & 4 & 3 & 7 \\ 5 & 8 & 1 & 8 \\ 7 & 6 & 9 & 4 \end{bmatrix}$$

b) Solve subset sum using backtracking: $S = \{5, 10, 12, 13, 15, 18\}$, $d = 30$. [8 Marks]

Part IV**SOFTWARE ENGINEERING & PROJECT MANAGEMENT****10 MSE-I Examinations****10.1 CS2103-1 MSE-I – February 2025**

Mid Semester Examination - I, February 2025

Duration: 1 Hour — Max Marks: 15

CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

- Q1.** a) List and explain attributes of a good software. [6 Marks]
b) Explain Incremental Development with diagram. [6 Marks]
Q2. a) Outline ACM/IEEE Code of Ethics. [6 Marks]
b) Explain Boehm's Spiral Model with diagram. [6 Marks]
Q3. a) Outline Structure of Requirements Document. [6 Marks]
b) With diagram explain types of Non-Functional Requirements. [6 Marks]
Q4. a) Explain Requirements Elicitation and Analysis Process with diagram. [6 Marks]
b) Develop Scenario for collecting Medical history in MHC-PMS. Draw Use cases. [6 Marks]
-

10.2 CS2103-1 MSE-I – February 2024

Mid Semester Examination - I, February 2024

Duration: 1 Hour — Max Marks: 15

CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

- Q1.** a) Discuss ethical issues in professional environment. Explain types of software applications. [6 Marks]
b) Explain spiral software process model with diagram. [6 Marks]
Q2. a) Mention essential attributes of good software. Explain SRS document structure. [6 Marks]
b) Explain incremental process model. Differences between waterfall and incremental. [6 Marks]
Q3. a) Define requirement. Discuss types of non-functional requirements with examples. [6 Marks]
b) Explain need of software engineering discipline with justifications. [6 Marks]
Q4. a) List readers of system requirements. Explain user and system requirements for MHC-PMS. [6 Marks]
b) Discuss requirement elicitation and analysis activity in RE process. [6 Marks]
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10.3 21CS404 MSE-I – June 2023

Mid Semester Examination - I, June 2023

Duration: 1 Hour — Max Marks: 20

21CS404 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

- Q1.** a) Analyze essential attributes of good software. [8 Marks]
 b) Explain various types of non-functional requirements. [8 Marks]
Q2. a) Discuss Waterfall model and give two advantages. [8 Marks]
 b) Explain different types of applications that may be developed. [8 Marks]
Q3. a) Analyze areas where standards not bound by laws but by professional responsibility. [8 Marks]
 b) Illustrate Boehm's spiral model. [8 Marks]
Q4. a) Build structure of requirements document. [8 Marks]
 b) Show process model of elicitation and analysis process. [8 Marks]

11 MSE-II Examinations

11.1 21CS404 MSE-II – July 2023

Mid Semester Examination - II, July 2023

Duration: 1 Hour — Max Marks: 20

21CS404 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

- Note:** 1) Answer any One full question from each Unit.
 2) Provide necessary diagrams wherever needed

Unit – I

- 1.** a) With neat diagram explain the organization of the Model-View-Controller. [5 Marks, L*2, CO* 2, PO* 1,2]
 b) Explain the context model with a neat diagram. [5 Marks, L2, CO* 2, PO* 1]
2. a) Draw and explain the sequence diagram for describing data collection for Weather Information System. [5 Marks, L2, CO* 3, PO* 1]
 b) Develop an architecture for a language processing system. [5 Marks, L3, CO* 2, PO* 1]

Unit – II

- 3.** a) Construct a state diagram of a microwave oven system. [5 Marks, L3, CO* 2, PO* 1,2]
 b) List and explain the principles of the agile method. [5 Marks, L2, CO* 3, PO* 1]
4. a) Briefly explain the five stages of the object-oriented design process. [5 Marks, L2, CO* 3, PO* 1]
 b) Explain the repository model with a neat block diagram. [5 Marks, L2, CO* 2, PO* 1]

12 SEE (Semester End Examinations)

12.1 CS2103-1 SEE – May 2025

Semester End Examination, May 2025

Duration: 3 Hours — Max Marks: 100

CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

- Q1.** a) Briefly explain essential attributes of good software. List fundamental SE activities. [8 Marks]
 b) Explain Incremental Development with diagram. [8 Marks]
Q2. a) Outline ACM/IEEE Code of Ethics. [8 Marks]
 b) Explain Boehm's Spiral Model with diagram. [8 Marks]
Q3. a) Outline Structure of Requirements Document. [8 Marks]
 b) With diagram explain types of Non-Functional Requirements. [8 Marks]
Q4. a) Explain Requirements Elicitation and Analysis Process with diagram. [8 Marks]
 b) Develop Scenario for MHC-PMS medical history. Draw Use cases. [8 Marks]
Q5. a) Briefly explain System Testing. [8 Marks]
 b) Describe five key UML diagram types with purposes. [8 Marks]
Q6. a) Briefly explain concept of system models. [8 Marks]
 b) Explain language processing system architecture. [8 Marks]
Q7. a) Briefly explain four test strategies for software development. [8 Marks]
 b) Explain ATM System software architecture in Transaction Processing Systems. [8 Marks]
Q8. a) Explain project scheduling concept and significance. [8 Marks]
 b) Explain configuration management concept. [8 Marks]
Q9. a) List and explain key stages in component measurement process. [8 Marks]
 b) Describe various risk identification in project management. [8 Marks]

12.2 CS2103-1 SEE – May 2024

Semester End Examination, May 2024

Duration: 3 Hours — Max Marks: 100

CS2103-1 – SOFTWARE ENGINEERING & PROJECT MANAGEMENT

Part B: Descriptive Questions

- Q1.** a) Explain software development life cycle. [8 Marks]
 b) Differentiate Waterfall model and Spiral model. [8 Marks]
Q2. a) Explain V-model in plan-driven software process. [8 Marks]
 b) Explain Boehm's spiral model. [8 Marks]
Q3. a) Distinguish functional and non-functional requirements. Explain types with diagram. [8 Marks]
 b) Explain incremental model. Advantages and disadvantages. [8 Marks]
Q4. a) Explain data flow model with insulin pump example. [8 Marks]
 b) Explain acceptance testing process stages with diagram. [8 Marks]
Q5. a) Explain design models: (i) Sequence models (ii) Use-case models. [8 Marks]
 b) Explain extreme programming and practices. [8 Marks]
Q6. a) Identify Weather Station object classes. Draw class diagrams. [8 Marks]

- b) What are agile methods? Describe principles. [8 Marks]
- Q7.** a) What is Risk and Risk management? Explain categories. [8 Marks]
- b) Explain project planning process with diagram. [8 Marks]
- Q8.** a) Explain phases of software review process with diagram. [8 Marks]
- b) Explain software product metrics and types. Describe static metrics. [8 Marks]

Part V**MICROPROCESSOR AND EMBEDDED SYSTEMS****13 MSE-I Examinations****13.1 CS3005-1 MSE-I – February 2024**

Mid Semester Examination - I, February 2024

Duration: 1 Hour — Max Marks: 15

CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Part B: Descriptive Questions

- Q1.** a) Explain internal architecture of 8086 microprocessor with diagram. [6 Marks]
b) Explain different addressing modes of 8086 with examples. [6 Marks]
Q2. a) Explain memory segmentation in 8086. [6 Marks]
b) Write ALP to find sum of array elements. [6 Marks]
-

13.2 21CS403 MSE-I – June 2023

Mid Semester Examination - I, June 2023

Duration: 1 Hour — Max Marks: 20

21CS403 – MICROPROCESSOR AND MICROCONTROLLERS

Part A: Multiple Choice Questions ($4 \times 1 = 4$ Marks)

1. The 8086 microprocessor has ___ bit data bus: (A) 8 (B) 16 (C) 20 (D) 32
2. Which register is used as loop counter in 8086: (A) AX (B) BX (C) CX (D) DX
3. Physical address in 8086 is calculated as: (A) Segment $\times$ 16 + Offset (B) Segment + Offset $\times$ 16 (C) Segment + Offset (D) Segment $\times$ Offset
4. Flag set when result is zero: (A) Carry Flag (B) Zero Flag (C) Sign Flag (D) Overflow Flag

Part B: Descriptive Questions

- Q1.** a) Explain internal architecture of 8086 with diagram. [8 Marks]
b) Explain different addressing modes of 8086 with examples. [8 Marks]
Q2. a) Write 8086 ALP to find largest element in array of 10 bytes. [8 Marks]
b) Explain interrupt structure of 8086 microprocessor. [8 Marks]

14 MSE-II Examinations**14.1 21CS403 MSE-II – July 2023**

Mid Semester Examination - II, July 2023

Duration: 1 Hour — Max Marks: 20

21CS403 – MICROPROCESSOR AND EMBEDDED SYSTEM

Note: Answer any One full question from each Unit.

Unit – I

1. a) Write the procedure to convert two ASCII values to the equivalent hexadecimal. Write appropriate comments. [4 Marks, L*2, CO* 2, PO* 1]

b) Develop an assembly language program to extract the hour from the system time and display in decimal at row number 8 and column number 20 in the format: HOUR: 10 where 10 is the current hour extracted from the system time. Write appropriate comments. [6 Marks, L3, CO* 2, PO* 2]

2. a) Explain the interrupts required to perform the following tasks by specifying the necessary parameters involved:

i. Buffered keyboard input

ii. Set system date

iii. Get cursor position

iv. Create sub directory

[4 Marks, L2, CO* 2, PO* 1]

b) Develop an assembly language program to read a string from the keyboard and write the same to a file. Display the appropriate error messages. Write appropriate comments. [6 Marks, L3, CO* 2, PO* 2]

Unit – II

3. a) Bring out the various differences between RISC and CISC machines. [5 Marks, L2, CO* 3, PO* 1]

b) Develop an embedded C program to display a message using scrolling effect on LCD for a suitable period of time in both directions. Do not use the built-in function for scrolling. [5 Marks, L3, CO* 4, PO* 2]

4. a) Explain the embedded system software in detail. [5 Marks, L2, CO* 3, PO* 1]

b) Develop an embedded C program to perform up-down counter on a 7-segment display using switches. [5 Marks, L3, CO* 4, PO* 2]

15 SEE (Semester End Examinations)**15.1 CS3005-1 SEE – May 2025**

Semester End Examination, May 2025

Duration: 3 Hours — Max Marks: 100

CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Part B: Descriptive Questions

Q1. a) Explain 8086 architecture with Bus Interface Unit and Execution Unit. [8 Marks]

b) Explain memory segmentation in 8086. [8 Marks]

Q2. a) Explain different addressing modes of 8086 with examples. [8 Marks]

b) Explain instruction set of 8086. [8 Marks]

-
- Q3.** a) Write ALP for string operations. [8 Marks]
b) Write ALP for array processing. [8 Marks]
Q4. a) Write ALP for arithmetic operations. [8 Marks]
b) Explain interrupt handling in 8086. [8 Marks]
Q5. a) Explain 8255 PPI. [8 Marks]
b) Explain timer/counter operations. [8 Marks]
-

15.2 CS3005-1 SEE – May 2024

Semester End Examination, May 2024

Duration: 3 Hours — Max Marks: 100

CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Part B: Descriptive Questions

- Q1.** a) Explain microprocessor evolution. [8 Marks]
b) Explain 8086 pin diagram. [8 Marks]
Q2. a) Explain Bus Interface Unit. [8 Marks]
b) Explain Execution Unit. [8 Marks]
Q3. a) Explain memory organization. [8 Marks]
b) Explain stack operations. [8 Marks]
Q4. a) Explain procedure calls. [8 Marks]
b) Explain interrupts. [8 Marks]
Q5. a) Explain DMA. [8 Marks]
b) Explain serial communication. [8 Marks]

Part VI**INNOVATION AND DESIGN THINKING****16 MSE-I Examinations****16.1 ME1654-1 MSE-I – February 2025**

Mid Semester Examination - I, February 2025

Duration: 1 Hour — Max Marks: 15

ME1654-1 – INNOVATION AND DESIGN THINKING

Part A: Multiple Choice Questions ($3 \times 1 = 3$ Marks)

1. Design Thinking approach primarily focuses on:
 - (A) Following strict rules and guidelines
 - (B) Understanding user needs and creating innovative solutions
 - (C) Minimizing team collaboration
 - (D) Avoiding experimentation
2. Which best describes “Empathy” in Design Thinking?
 - (A) Finding faults in user behavior
 - (B) Understanding users’ feelings, thoughts, and motivations
 - (C) Designing based on assumptions
 - (D) Testing solutions without user involvement
3. The “Define” stage is primarily about:
 - (A) Generating as many ideas as possible
 - (B) Clearly articulating the problem based on insights from empathy
 - (C) Building and testing prototypes
 - (D) Observing users in their natural environment

Part B: Descriptive Questions ($2 \times 6 = 12$ Marks)**UNIT – I**

- Q1.** a) Explain the five stages of Design Thinking process with examples. [6 Marks]
b) What is the importance of empathy in Design Thinking? Explain methods to build empathy with users. [6 Marks]

UNIT – II

- Q2.** a) Describe the Ideation phase. Explain any two ideation techniques in detail. [6 Marks]
b) What is prototyping? Explain different types of prototypes used in Design Thinking. [6 Marks]

17 SEE (Semester End Examinations)**17.1 ME1654-1 SEE – May 2025**

Semester End Examination, May 2025

Duration: 3 Hours — Max Marks: 100

ME1654-1 – INNOVATION AND DESIGN THINKING

Part B: Descriptive Questions

- Q1.** a) Explain Design Thinking process in detail. [8 Marks]
b) Explain Empathize stage methods. [8 Marks]
Q2. a) Explain Define stage techniques. [8 Marks]
b) Explain Ideate stage tools. [8 Marks]
Q3. a) Explain different Prototype types. [8 Marks]
b) Explain Test stage approaches. [8 Marks]
Q4. a) Explain innovation frameworks. [8 Marks]
b) Explain creativity techniques. [8 Marks]
Q5. a) Explain user research methods. [8 Marks]
b) Discuss design challenges. [8 Marks]

END OF QUESTION BANK

This comprehensive document contains all examination papers:

MSE-I: February 2025, February 2024, June 2023

MSE-II: April 2025, July 2023, June 2022

SEE: May 2025, May 2024, July 2024, January 2023, August 2023

Supplementary: September 2022

Subjects: Linear Algebra, DBMS, DAA, SEPM, Microprocessor, Design Thinking

Comprehensive Exam Question Bank

NMAM Institute of Technology, Nitte
Descriptive Questions Only (No MCQs)

B.Tech (CBCS) — IV Semester

Compiled from SEE & MSE Papers (2022–2025)

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Software Engineering and Project Management

CS2103-1 SEE — May 2025

1. a) Briefly explain the essential attributes of a good software. List out the fundamental software engineering activities.
b) Explain Incremental Development with a diagram.
2. a) Outline ACM/IEEE Code of Ethics.
b) Explain Boehm's Spiral Model with diagram.
3. a) Outline the Structure of Requirements Document.
b) With diagram explain the types of Non-Functional Requirements.
4. a) Explain Requirements Elicitation and Analysis Process with diagram.
b) Develop the Scenario for collecting Medical history in MHC-PMS and draw Use cases for the MHC-PMS.
5. a) Briefly explain the System Testing.
b) Describe the five key UML diagram types used in system modeling with their purposes.
6. a) Briefly explain the concept of system models.
b) Explain the architecture and working of language processing system in architectural design.
7. a) Briefly explain any four different test strategies used for conventional software development.
b) Explain the Software Architecture of an ATM System used in Transaction Processing Systems.
8. a) Explain the concept of project scheduling and its significance in project management.
b) Explain the concept of configuration management in software development projects.
9. a) List and explain the key stages in component measurement process.
b) Describe various risk identification in project management.

CS2103-1 SEE — May 2024

1. a) Explain software development life cycle.
b) Differentiate between Waterfall model and Spiral model of software development.
2. a) Explain the V-model in plan-driven software process.
b) Explain Boehm's spiral model.
3. a) Distinguish between functional and non-functional requirements. With a block diagram, briefly explain non-functional requirement types.
b) Explain incremental model. Highlight its advantages and disadvantages.
4. a) Explain data flow model with an example of insulin pump.
b) Explain with a neat diagram, stages in the acceptance testing process.

5. a) Explain the following design models with a simple example for each: (i) Sequence models (ii) Use-case models.
b) Explain extreme programming and its practices.
6. a) Identify the various Weather Station object classes and draw their respective class diagrams.
b) What are agile methods? Describe the principles of agile method.
7. a) What is Risk and Risk management? Explain with suitable examples, the categories of risks.
b) Explain the project planning process with a neat diagram.
8. a) Explain with a help of neat diagram, the phases of software review process.
b) Explain software product metrics and its types. Describe the various static software product metrics.

AD1103-1 SEE — May 2024

1. a) Briefly explain the essential attributes of good software. List out the fundamental software engineering activities.
b) Explain the Waterfall model. What are the advantages and disadvantages?
2. a) Briefly, discuss various phases of software development life cycle.
b) What is scrum? Briefly explain the key ceremonies a scrum team might partake in.
3. a) Explain the structure of a requirements document in requirement engineering.
b) Explain the principles of agile methods.
4. a) Briefly explain the System Testing.
b) You have been asked to develop a system that will help with planning large-scale events and parties such as weddings, graduation celebrations, birthday parties, etc. Using an activity diagram, model the process context for such a system.
5. a) How to identify object classes in object-oriented systems? Briefly explain the Architectural design using the use case of Weather station.
b) Briefly explain the integration and smoke testing Strategies for Conventional Software.
6. a) Briefly explain the concept of system models.
b) Explain the four fundamental architectural views.
7. a) Describe the importance of risk identification in project management.
b) Explain the concept of configuration management in software development projects.
8. a) Explain the concept of project scheduling and its significance in project management.
b) Discuss the importance of quality management in project execution and delivery.

AD1103-1 SEE — May 2025

1. a) Explain the structure of a requirements document in requirement engineering.
b) Explain the various checks involved in requirement validation in detail.
c) Explain the different types of interviews conducted during the requirement collection process in software development.
2. a) Briefly, discuss various phases of software development life cycle.
b) Briefly explain the essential attributes of good software. List out the fundamental software engineering activities.
3. a) Explain the Waterfall model. What are the advantages and disadvantages?
b) Explain the working of the Scrum framework in software development. Discuss the key activities involved in the Scrum process.
4. a) Briefly explain the System Testing.
b) Describe the five key UML diagram types used in system modeling with their purposes.
5. a) Briefly explain the concept of system models.
b) Explain the architecture and working of language processing system in architectural design.
6. a) Briefly explain any four different test strategies used for conventional software development.
b) Explain the Software Architecture of an ATM System used in Transaction Processing Systems.
7. a) Explain the concept of project scheduling and its significance in project management.
b) Explain the concept of configuration management in software development projects.
8. a) List and explain the key stages in component measurement process.
b) Describe various risk identification in project management.

19IS8X83 SEE — May-June 2023

1. a) Define Software Engineering. What are the different types of software applications?
b) Explain the incremental development process model with a neat block diagram. List its advantages and disadvantages.
2. a) Define requirements specification. Explain different ways of writing a system requirement specification.
b) Explain the different attributes of a good software.
3. a) Explain waterfall model of software development process. What are its advantages and disadvantages?
b) What is requirement validation? Explain the checks and techniques followed for process of requirement validation.
4. a) Explain the general model of design process.
b) Summarize the different sectors into which each loop in Boehm's spiral model is split.
5. a) Explain the state diagram of a microwave oven.

- b) Explain the activities to be carried out in an object-oriented design process of a system.
6.
 - a) Illustrate sequence diagram with an example to view patient information.
 - b) Explain the principles of agile methods.
 - c) Explain the organization of MVC pattern with a neat diagram.
7.
 - a) Explain transaction processing system with a neat diagram showing software architecture of an ATM system.
 - b) Explain generalization with an example.
 - c) Summarize the scrum process of agile project management.
8.
 - a) What is risk management? Outline the risk management process with a neat diagram and list out the types of risk in a risk checklist.
 - b) What are the factors influencing the effectiveness and efficiency of communications in a group?
 - c) Explain the factors affecting software pricing.
9.
 - a) Explain the workflow for a project planning process with a neat diagram.
 - b) Explain the three phases of software review process.
 - c) What are the benefits of creating a cohesive group?

CS2103-1 MSE-I — February 2025

1.
 - a) List and explain the attributes of a good software.
 - b) Explain Incremental Development with diagram.
2.
 - a) Outline ACM/IEEE Code of Ethics.
 - b) Explain Boehm's Spiral Model with diagram.
3.
 - a) Outline the Structure of Requirements Document.
 - b) With diagram explain the types of Non-Functional Requirements.
4.
 - a) Explain Requirements Elicitation and Analysis Process with diagram.
 - b) Develop the Scenario for collecting Medical history in MHC-PMS and draw Use cases for the MHC-PMS.

CS2103-1 MSE-I — February 2024

1.
 - a) Discuss the ethical issues that might arise in a professional environment. Explain the various types of software applications.
 - b) Explain with a neat diagram spiral software process model.
2.
 - a) Mention the essential attributes of a good software. Also, explain the structure of a software requirements specification (SRS) document for any software system of your choice.
 - b) Explain incremental software process model with a suitable diagram. Also, briefly explain the key differences between waterfall and incremental process model.
3.
 - a) Define requirement. Also, discuss various types of non-functional requirements with suitable example for any software system of your choice.

- b) Explain the need of software engineering discipline in a modern era with suitable justifications.
- 4.
 - a) List the readers of system requirements. Also, explain user and system requirements along with suitable examples for mental healthcare patient monitoring software (MHC-PMS) system.
 - b) Discuss briefly with suitable example the requirement elicitation and analysis activity in requirements engineering (RE) process.

21CS404 MSE-I — June 2023

- 1.
 - a) Analyze the essential attributes of a good software.
 - b) Explain various types of non-functional requirements.
- 2.
 - a) Discuss Waterfall model software process and give any two of its advantages.
 - b) Explain the different types of application that may be developed.
- 3.
 - a) Analyze the areas where standards of acceptable behavior are not bound by laws but by the more tenuous notion of professional responsibility.
 - b) Illustrate the Boehm's spiral model of the software process.
- 4.
 - a) Build the structure of a requirements document.
 - b) Show a process model of the elicitation and analysis process.

Database Management Systems

CS2102-1 SEE — May 2025

1.
 - a) What are the Characteristics of Database approach? Explain.
 - b) With a neat diagram, explain the three Schema architecture.
2.
 - a) Explain the following types of attributes with examples: (i) Composite (ii) Multivalued (iii) Atomic.
 - b) Explain the Cardinality Ratio for Binary relationships with examples.
3.
 - a) Consider the following Tables:
STUDENT(SName, SUSN, SPhoneNo, SAddress, SAge, SCGPA)
COURSE (Ccode, Ctitle, SUSN, Sname, Cstaff)
(i) What is the Degree of the relation STUDENT?
(ii) What is the Degree of the relation COURSE?
(iii) Construct a Relational algebra to display all details of students with SCGPA>8.0.
(iv) Construct a Relational algebra to display names of all students who have taken up 'DBMS' course.
(v) Construct a relational algebra to display the names of all courses taken up by students with SCGPA>9.0.
 - b) Explain the steps in ER to Relational Mapping Algorithm.
4.
 - a) Explain how you can Specify the Not Null, DEFAULT and CHECK constraints in SQL.
 - b) Explain the CASCADE and RESTRICT options of DROP TABLE command.
 - c) What are the uses of ALTER TABLE command?
5.
 - a) What are the Informal Design Guidelines for relational schema? Mention all and explain any 2.
 - b) What is a Functional Dependency? Explain with an example.
6.
 - a) Explain the Standard Inference rules for Functional Dependencies.
 - b) Explain the algorithm for determining X^+ , the Closure of X under F .
7.
 - a) What is a B+ Tree? Explain structure & characteristics of a B+ Tree with a neat diagram.
 - b) Briefly explain the ACID properties.
8.
 - a) What is a Query Optimizer? Explain its working with a neat diagram.
 - b) What are the 2 rules of Strict 2PL? Briefly explain.

CS2102-1 SEE — July 2024

1.
 - a) Briefly explain the various characteristics of the database approach.
 - b) Explain the following Relational Algebra operations with example for each: (i) Division (ii) Union (iii) Cartesian product (iv) COUNT aggregate function.
2.
 - a) Design an ER diagram and schema for the flight management system. Specify any assumptions made.
 - b) Explain grouping concepts in relational algebra. Give one example.

- c) Explain following aggregate functions used in relational algebra. Give example for each: (i) AVERAGE (ii) COUNT.
3.
 - a) With a neat block diagram explain the three schema architecture for databases.
 - b) Consider the Sailors-Boats-Reserves DB: SAILOR(sid, sname, rating, age), BOAT(bid, bname, color), RESERVES(sid, bid, date). Write each of the following queries in Relational Algebra:
 - (i) Find all sailor id's of sailors who have a rating of at least 8 or reserved boat 103.
 - (ii) Find the names of sailors who have reserved a red and a green boat.
 - (iii) Find the sids of sailors with age over 20 who have not reserved a red boat.
4.
 - a) Consider the relation schema: Employee(employee-name, street, city), Works(employee-name, company-name, salary), Company(company-name, city), Manages(employee-name, manager-name). Answer the following queries:
 - (i) Find the names of all employees in the database who do not work for 'First Bank Corporation'.
 - (ii) Find the names of all employees in the database who earn more than every employee of 'Small Bank Corporation'.
 - b) What is a view? Give the specification of view in SQL.
5.
 - a) Consider $R = (A, B, C, D, E, G)$ and FD set $F = \{AB \rightarrow C, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, BC \rightarrow A, E \rightarrow G\}$. Discover whether the following binary decompositions have lossless join property:
 - (i) $R_1 = \{A, B, C\}$ and $R_2 = \{A, C, D, E, G\}$
 - (ii) $R_1 = \{A, B, C, D, E\}$ and $R_2 = \{A, D, G\}$
 - b) How basic constraints are specified in SQL-99. Explain with example. Give syntax.
6.
 - a) Consider the following relations for an order processing database: CUSTOMER(cust#, cname, city), ORDER(order#, odate, cust#, ord-Amt), ORDER-ITEM(order#, item#, qty), ITEM(item#, unit price), SHIPMENT(order#, warehouse#, ship-date), WAREHOUSE(warehouse#, city). Develop SQL queries for:
 - (i) Produce a listing: CUSTNAME, #oforders, AVG_ORDER_AMT.
 - (ii) For each item that has more than two orders, list the item, number of orders shipped from at least two warehouses and total quantity.
 - b) What is the need for normalization? Explain first, second and third normal form by giving example for each.
7.
 - a) Explain Strict Two-Phase Locking protocol in detail.
 - b) Explain the following index data structures in detail: (i) Hash based indexing (ii) Tree based indexing.
8.
 - a) Illustrate the anomalies associated with Interleaved Execution. Explain in detail.
 - b) Give the algorithm to insert an element in a B+ tree.

20CS502 SEE — January 2023

1.
 - a) Briefly explain the various characteristics of the database approach.
 - b) Design an ER diagram and schema for the flight management system. Specify any assumptions made.
2.
 - a) With a neat flow diagram explain the main phases involved in the database design process.

- b) Explain the following with example for each: (i) Structural constraints (ii) Recursive relationships.
3.
 - a) Consider the Sailors-Boats-Reserves DB: SAILOR(sid, sname, rating, age), BOAT(bid, bname, color), RESERVES(sid, bid, date). Write queries in Relational Algebra:
 - (i) Find all sailor id's of sailors who have a rating of at least 8 or reserved boat 103.
 - (ii) Find the names of sailors who have reserved a red and a green boat.
 - (iii) Find the sids of sailors with age over 20 who have not reserved a red boat.
 - b) Illustrate the following relational algebra operations with example for each: (i) Division (ii) Natural join.
 - c) Explain any 4 aggregate functions used in relational algebra. Give example for each.
4.
 - a) Consider the database schema: Employee(name, ssn, address, sex, salary, superssn), Project(pname, pnumber, plocation), Works_On(essn, pno, hours). Specify in relational algebra:
 - (i) Retrieve the names of all female employees who do not work on any project.
 - (ii) Find the names of all employees who are directly supervised by 'Franklin Wong'.
 - (iii) For each project on which more than 5 employees work, list the project name and the total hours per week spent on that project.
 - b) Explain the ER-to-Relational Mapping Algorithm with example.
 - c) Explain grouping concepts in relational algebra. Give one example.
5.
 - a) Consider the banking database: BRANCH, ACCOUNT, DEPOSITOR, CUSTOMER, LOAN, BORROWER. Write SQL queries:
 - (1) Find all the customers who have at least 2 accounts at all branches located in a specific city.
 - (2) Find all the customers who have accounts in at least 1 branch located in all the cities.
 - (3) Find all the customers who have accounts in at least 2 branches located in a specific city.
 - b) How basic constraints are specified in SQL-99. Explain with example. Give syntax.
6.
 - a) Consider the student enrollment database: STUDENT, COURSE, ENROLL, BOOK_ADOPTION, TEXT. Write SQL queries:
 - (i) Produce a list of text books in alphabetical order for courses offered by the 'CS' department that use more than two books.
 - (ii) List any department that has all its adopted books published by a specific publisher.
 - (iii) List the bookISBNs and book titles of the department that has maximum number of students.
 - b) What is a view? Give the specification of view in SQL.
7.
 - a) Consider the two sets of FDs: $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$. Check whether they are equivalent.
 - b) List and explain the various informal design Guidelines for Relation Schemas.
8.
 - a) What is the need for normalization? Explain first, second and third normal form by giving example for each.
 - b) Consider $R = \{A, B, C, D, E, F, G, H\}$ and FDs: $F = \{A \rightarrow B, ABCD \rightarrow E, EF \rightarrow G, EF \rightarrow H, ACDF \rightarrow EG\}$. Find a minimal cover G for F .
9.
 - a) Explain Strict Two-Phase Locking protocol in detail.

- b) With an example, Explain aggregate data models of NoSQL.
- 10. a) Illustrate the anomalies associated with Interleaved Execution. Explain in detail.
- b) What is CAP theorem? Explain in detail.

CS2102-1 MSE-I — February 2025

1. a) Design and construct an ER diagram for “Student Database Management System”. Indicate all keys, constraints and assumptions that are made.
- b) Discuss the main characteristics of the database approach.
2. a) Identify the Entity, Relationship, Participation Constraint and Cardinality ratio for the ER diagram of Movie database schema along with the justification. (ER diagram with ACTOR, MOVIE, DIRECTOR, PRODUCER entities with PERFORMS_IN, LEAD_ROLE, ALSO_A_DIRECTOR, DIRECTS, PRODUCES relationships)
- b) Describe Three Schema Architecture with a neat diagram.
3. a) Write the following queries in Relational Algebra expression for the given database:
 EMPLOYEE(SSN, SNAME, AGE, SALARY, DEPT_ID)
 DEPARTMENT(DEPT_ID, DEPT_NAME, DEPT_HEAD)
 PROJECT(PID, DEPT_ID, PNAME)
 WORKSON(SSN, PID, PHOURS)
 (i) Get the names of employees working on projects managed by the ‘HR’ department.
 (ii) Retrieve the names of employees who do not work on any project.
 (iii) Find the names of all departments that have at least one project.
- b) Explain how to deal with constraints violation during insertion and delete operations specified in the relational model.
4. a) Explain ER to relational mapping for the given ER diagram (Bank–Branches–Bank-Branch, Account, Loan, Customer with ACCTS, LOANS, A-C, L-C relationships).
- b) List and explain characteristics of Relations.

CS2102-1 MSE-II — April 2025

1. a) Consider the following schema and write the SQL queries:
Suppliers(sid: integer, sname: string, address: string)
Parts(pid: integer, pname: string, color: string)
Catalog(sid: integer, pid: integer, cost: double)
 (i) Find the sids of suppliers who supply some red or green part.
 (ii) Find the sids of suppliers who supply only red parts.
 (iii) Find the pnames of parts for which there is some supplier.
 (iv) Find the sids of suppliers who supply some red part or are located at 221 Packer Street. (solution should be in Nested Query)
2. a) Consider the following schema and write SQL queries:
Movie(movie_id, title, release_year, genre, director_id, rating)
Director(director_id, name, birth_year, nationality)
Actor(actor_id, name, birth_year, nationality)
Movie_Actor(movie_id, actor_id, role)
Review(rev_id, user_id, movie_id, rev_text, rating, rev_date)
User(user_id, username, email, age)
 (i) Find the names of all actors who have acted in at least one movie released after

2015.

- (ii) Find the directors who have directed more than 5 movies.
 - (iii) Get the average rating of all movies directed by 'Christopher Nolan'.
 - (iv) Find movies that have a higher rating than the average rating of movies released in the same genre. (solution should be in Nested Query)
3. a) Consider the two sets of functional dependencies $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$. Are they equivalent?
b) What is Normalization? Explain the first Normal form with example.
 4. a) Suppose a relational schema $R(A B C D E F G H I J)$ and set of functional dependencies $X: \{(A, B) \rightarrow C, A \rightarrow \{D, E\}, B \rightarrow F, F \rightarrow \{G, H\}, D \rightarrow \{I, J\}\}$. What is candidate key of R ? Check out whether the relation is in 3NF or not. If not decompose it in 3NF.
b) Explain any two Informal Design Guidelines for Relation Schemas.

CS2102-1 MSE-I — February 2024

1. a) Describe with a neat diagram of Three Schema Architecture. Why do we need mappings among schema levels?
b) Design and construct an ER diagram for "Engineering College Management System". Indicate all keys, constraints and assumptions that are made.
2. a) Discuss the main characteristics of database approach.
b) Explain how domain constraints and referential integrity constraints are specified in relational model.
3. a) Define an Entity attribute. Explain different types of attribute that occur in an E-R diagram model with an example.
b) Outline the steps to convert the basic E-R model to relational database scheme.
4. a) "For a good database design relation schema should be such that matching attributes must be primary key or foreign key otherwise it leads to spurious tuples". Justify.
b) Discuss the characteristics of relations.

Design and Analysis of Algorithms

CS3004-1 SEE — May 2025

1. a) Explain the Performance Analysis of an algorithm.
b) Write the pseudocode to sort the elements in an array using selection sort and analyze its efficiency. Apply selection sort to sort the following elements in an array: 5, 4, 2, 1, 6, 3.
2. a) Write the general plan for analyzing the efficiency of non-recursive algorithm. Write the pseudocode to find the maximal element in an array and analyze its efficiency.
b) Write Divide And Conquer recursive Merge sort algorithm and derive the time complexity of this algorithm.
3. a) Write the algorithm for brute force string matching. Show the comparisons the Brute force string matching makes for the pattern $P = \text{acbcac}$ in the text $T = \text{acbcababcaacbcac}$.
b) Illustrate Quick sort algorithm. Sort the following list using Merge sort: 38, 27, 43, 3, 9, 82, 10.
4. a) Write an insertion sorting algorithm using decrease and conquer technique. Apply the algorithm on the following numbers and trace the steps: 23, 1, 10, 5, 2.
b) Construct AVL tree for the following input by showing each step of the insertion process and type of rotation used for transformation: 63, 9, 19, 27, 18, 108, 99, 81.
5. a) Identify the pattern “BARBER” in the given text “JIM_SAW_ME_IN_A_BARBERSHOP” using Horspool’s string-matching algorithm. Show the construction of shift table for the same.
b) Write an algorithm for sorting list of integers by comparison method that uses input enhancement design technique. Apply the algorithm on the following numbers and trace the steps: 25, 45, 10, 20, 50, 15.
6. a) Define topological sorting. Apply DFS based method to topologically sort the vertices of a graph (with vertices 0,1,2,3,4 and directed edges).
b) Write and Apply Floyd’s all pairs shortest path algorithm for a given graph (with vertices 1,2,3,4 and weighted edges).
7. a) Solve the given instance of sum of subset problem $S = \{5, 10, 12, 13, 15, 18\}$ and $d = 30$. Construct a state space tree.
b) Define minimum cost spanning tree. Write Prim’s algorithm to find minimum cost spanning tree.
8. a) Explain the following: (i) Class P (ii) Class NP (iii) NP Complete Problem (iv) NP Hard Problem.
b) Solve the job assignment problem using Branch and Bound technique for the following matrix:

	Job 1	Job 2	Job 3	Job 4
A	9	2	7	8
B	6	4	3	7
C	5	8	1	8
D	7	6	9	4

CS3004-1 MSE-I — February 2025

1. a) Summarize the algorithm design and analysis process with a neat diagram.
 b) Consider the following algorithm:
 Algorithm fun($A[0 \dots n-1]$)
 $x \leftarrow A[0]$
 for $i \leftarrow 1$ to $N - 1$ do
 if $A[i] < x$
 $x = A[i]$
 return x
 (i) What does this algorithm compute? (ii) What is the basic operation? (iii) How many times basic operation executed? (iv) What is the efficiency of this algorithm?
2. a) Define asymptotic notations Big Oh (O) and Big Theta (Θ) and give example.
 b) Show that the efficiency of Tower of Hanoi is exponential.
3. a) Develop an algorithm to find the existence of a pattern in a given string and mathematically analyze its worst-case complexity.
 b) Outline merge sort algorithm and derive its time complexity.
4. a) Define travelling salesman problem. Solve the following travelling salesman problem using exhaustive search technique and state its efficiency. (Graph with vertices A, B, C, D and weighted edges: A-B=6, A-C=3, A-D=1, B-C=4, B-D=2, C-D=8)
 b) The marks scored by five students in an exam are as follows: 8, 10, 5, 2, 14. Apply a quick sort recursive sorting algorithm to sort these marks in ascending order and show its tree of recursive call.

CS3004-1 MSE-II — April 2025

1. a) Outline Insertion sort algorithm and show the sorting of list E, X, A, M, P, L, E in alphabetical order using insertion sort method.
 b) Apply heap sort algorithm to sort the list 90, 89, 56, 72, 67, 12, 21, 34, 45.
2. a) Define AVL tree and construct AVL tree for the data {2, 9, 3, 12, 15, 8, 11}.
 b) Define topological sorting. Illustrate the topological sorting using Source Removal method for a given graph (with vertices and directed edges).
3. a) Write an algorithm to sort list of integers using Comparison Counting and compare its time complexity with the time complexity of selection sort and Bubble sort.
 b) Summarize the steps of Horspool string matching algorithm and search for the pattern TIME in the text A STITCH IN TIME SAVES NINE.
4. a) Apply All Pair Shortest path algorithm for a given graph (with vertices A, B, C, D, E and weighted edges).
 b) Write an algorithm for Breadth-First Search (BFS). List any two facts that differentiate BFS from DFS.

CS3004-1 MSE-I — February 2024

1. a) Explain the properties of an algorithm taking linear search as example.
 b) For each of the following functions, indicate how much the function value will change if its argument is increased fourfold: (i) $\log_2 n$ (ii) n (iii) n^2 (iv) 2^n (v) $\sqrt{n}$.

2. a) Explain the general framework for Analysis of an algorithm.
b) Define 3 asymptotic notations and determine whether the following assertions are true or false: (i) $10n^2 + 9 \in O(n^2)$ (ii) $n(n+1)/2 \in O(n^2)$.
3. a) Write the general plan for analyzing the efficiency of recursive algorithm. Write the pseudo code to find factorial of a number and analyze its efficiency.
b) Give two brute force algorithms to sort the given list of integers.
4. a) Illustrate Merge sort algorithm and analyze its efficiency. Sort the following list using Merge sort: 6, 14, 8, 2, 15, 10, 7, 5.
b) Show the comparisons the Brute force string matching makes for the pattern P=0001 in the text T=000010001010101.

21CS402 MSE-I — June 2023

1. a) With supporting process flow diagram explain algorithm design and analysis.
b) Design a recursive algorithm to solve $2^n = 2^{n-1} + 2^{n-1}$ and perform mathematical analysis to find the complexity of that algorithm.
2. a) Define three asymptotic notations and express $f(n) = n(n-1)/2$ using three asymptotic notations with proof by its definition.
b) With pseudo code perform complexity analysis of sequential search algorithm for the three types of efficiencies.
3. a) What is a brute force design technique? Give brute force string matching algorithm, perform complexity analysis and trace for the example: TEXT: JIM_SAW_ME_IN_SHOE SHOP, PATTERN: SHE.
b) Apply merge sort to sort the list L,A,S,T,S,O,R,T,I,N,G in alphabetical order. Perform complexity analysis of merge sort.
4. a) Give complete Quick sort algorithm and analyze it for complexity. Trace the algorithm for the given unsorted list: 5, 3, 1, 9, 8, 2, 4, 7.
b) What is an exhaustive search method? For a knapsack problem $n = 4$, capacity $m = 10$, $w = (4, 6, 3, 5)$ and values $= (30, 50, 20, 10)$, give solution using exhaustive search method.

21CS402 Makeup SEE — September 2023

1. a) Write an Algorithm to check whether all the elements in a given array are distinct. Derive its time efficiency.
b) Discuss Brute Force string matching Algorithm. What is the efficiency of this Algorithm in worst case?
c) Write an Algorithm to sort n numbers using Merge sort. Discuss its efficiency in worst case.
2. a) Design an Algorithm to search an element using sequential search. Discuss its efficiency in best, worst and average cases.
b) Show the complexity of selection sort by writing and analyzing an Algorithm.
c) Apply Quick sort to sort the following numbers in alphabetical order. Discuss the time complexity of quick sort in all three cases.
3. a) Discuss Big Oh and Big Theta asymptotic notations with suitable example.

- b) Design one recursive Algorithm to find the number of binary digits in a positive decimal number n . Show its complexity.
- c) Discuss general Divide and Conquer procedure. Write an Algorithm to search an element in an array using Binary search.

Linear Algebra and its Applications

MA2005-1 SEE — May 2025

1. a) Diagonalize the matrix $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$. Hence find A^6 .
 b) Use Gauss-Seidel iteration method to solve the following system of linear equations $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$, start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry out three iterations.
 c) Find the general solution of the system whose augmented matrix is $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$.
2. a) Find the value of a and b so that the equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + az = b$ has (i) no solution (ii) a unique solution (iii) more than one solution.
 b) Solve the system of equations $x + y + z = 1$, $4x + 3y - z = 6$, $3x + 5y + 3z = 4$ by LU factorization method.
 c) Using the power method, find the dominant eigen value and corresponding eigen vector of the matrix $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}$ starting with the initial value $[1 \ 0 \ 0]^T$. Carry out four iterations.
3. a) Find all the eigen values of $A = \begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}$. Find an eigen vector corresponding to the largest eigen value.
 b) Reduce the given matrix A to echelon form and find its rank: $A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$.
 c) Define Trace of a matrix. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$ then verify that (i) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ (ii) $\text{tr}(AB) = \text{tr}(BA)$.
4. a) Let $T : R^3 \rightarrow R^2$ be a linear map defined by $T(x, y, z) = (y - x, y - z)$. Find the null space and range space of T . Also find dimension of null space and range space of T .
 b) Find a basis for the row space of $\begin{bmatrix} 1 & -3 & 4 & -1 & 9 \\ -2 & 6 & -6 & -1 & -10 \\ -3 & 9 & -6 & -6 & -3 \\ 3 & -9 & 4 & 9 & 0 \end{bmatrix}$.
 c) Check whether the set of vectors $\{(1, 2, 3), (4, 5, 6), (2, 1, 0)\}$ is linearly dependent or linearly independent.
5. a) Prove that a nonempty subset W of a vector space V over the field F is a subspace if and only if for any $\alpha, \beta \in W$ and $c \in F$, $c\alpha + \beta \in W$.
 b) Let $W = \text{span}\{x_1, x_2, x_3\}$ where $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$. Construct an orthogonal basis for W using Gram-Schmidt process.
 c) Find the co-ordinate vector of $(2, 7, -4)$ in R^3 with respect to the basis $\{(1, 2, 0), (1, 3, 2), (0, 1, 3)\}$.

6. a) Determine whether $(1, 1, 1, 1)$, $(1, 2, 3, 2)$, $(2, 5, 6, 4)$, $(2, 6, 8, 5)$ form a basis of R^4 . If not find the dimension of the subspace they span.
- b) Prove that a set of vectors $\{v_1, v_2, \dots, v_n\}$ is linearly dependent if and only if at least one of them is linear combination of the others.
- c) Find the matrix of linear transformation $T : V_2(R) \rightarrow V_3(R)$ defined by $T(x, y) = (x + y, x, 3x - y)$ with respect to the bases $B_1 = \{(1, 1), (3, 1)\}$ and $B_2 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$.
7. a) Find QR factorization of matrix $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$.
- b) Find a least square solution of $AX = B$ for $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$.
8. a) Find a singular value decomposition of matrix $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$.
- b) Convert the matrix $\begin{bmatrix} 1 & 5 & 2 & 6 & 7 & 3 \\ 3 & 11 & 6 & 8 & 15 & 11 \end{bmatrix}$ of observations to mean deviation form and construct the sample covariance matrix.

MA2005-1 MSE-I — February 2025

1. a) Find the general solution of the linear system:
 $x_1 + 3x_2 - 5x_3 = 4$
 $x_1 + 4x_2 - 8x_3 = 7$
 $-3x_1 - 7x_2 + 9x_3 = -6$
- b) Find the LU factorization of the matrix $\begin{bmatrix} 2 & 4 & -1 & 5 & -2 \\ -4 & -5 & 3 & -8 & 1 \\ 2 & -5 & -4 & 1 & 8 \\ -6 & 0 & 7 & -3 & 1 \end{bmatrix}$.
2. a) Determine the value of a such that the following system in unknowns x, y, z has: (i) No solution (ii) More than one solution.
 $x + y - z = 1$
 $2x + 3y + az = 3$
 $x + ay + 3z = 2$
- b) Using Gauss Seidel method, find an approximate solution of the linear system $10x + y + z = 12$, $2x + 10y + z = 13$, $2x + 2y + 10z = 14$ using the initial guess $x^{(0)} = y^{(0)} = z^{(0)} = 0$. Perform three iterations.
3. a) Find the eigen values and their corresponding eigen vectors of the matrix $\begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.
- b) If $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 2 & 0 \end{bmatrix}$, verify that:
 (i) $\text{tr}(AB) = \text{tr}(BA)$
 (ii) $\text{tr}(AA^T) = \text{sum of the squares of the elements of } A$.

4. a) Diagonalize the matrix $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.
- b) Using Rayleigh's power method obtain an approximate value of the largest eigen value and its corresponding eigen vector of the matrix $\begin{bmatrix} 1 & 2 & 4 \\ 2 & -1 & 2 \\ 3 & 2 & -1 \end{bmatrix}$ starting with an initial approximation $X_0 = [1 \ 0 \ 0]^T$. Perform three iterations.

MA2005-1 MSE-II — April 2025

1. a) (i) Define a subspace of a vector space V over a field F .
(ii) If G and H are subspaces of a vector space V over F then prove that $G \cap H$ is also a subspace of V .
- b) Define (i) Row space (ii) Null space of a matrix. Check whether the vector $\begin{pmatrix} 3 \\ 3 \\ -4 \end{pmatrix}$ is in the column space of the matrix $A = \begin{pmatrix} 1 & -3 & 4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{pmatrix}$.
2. a) (i) Define Linearly dependent and Linearly independent of a set of vectors.
(ii) Express $u = (1, -2, 5)$ in R^3 as a linear combination of the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 3)$, $u_3 = (2, -1, 1)$.
- b) Find the dimension of the subspace W of R^3 spanned by the vectors $\{(2, 4, 2), (1, -1, 0), (1, 2, 1), (0, 3, 1)\}$. Extend the basis of W to a basis of R^3 .
3. a) (i) State Rank-Nullity theorem.
(ii) Let a linear transformation $T : R^3 \rightarrow R^3$ be defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Find the null space of T and range space of T . Also find dimension of the null space of T and dimension of range space of T .
- b) Check whether a transformation $T : R^2 \rightarrow R^2$ defined by $T(x, y) = (x + y, x)$ is a linear transformation or not.
4. a) Find the matrix of the linear transformation $T : R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - y + 2z, 3x + y)$ with respect to bases $B = \{(1, 1, 1), (1, 2, 3), (1, 0, 0)\}$ and $B' = \{(1, 1), (1, -1)\}$.
- b) Let V and W be vector spaces over the field F . Let T and U be linear transformation from V into W . Then prove that the function $(T + U)$ defined by $(T + U)\alpha = T(\alpha) + U(\alpha)$ is a linear transformation from V into W . Also prove, if c is any element of F , the function (cT) defined by $(cT)(\alpha) = c(T\alpha)$ is a linear transformation from V into W .

MA2005-1 MSE-I — February 2024

1. a) Solve the following system of equations and obtain the general solution:
 $x_1 + 3x_2 - 5x_3 = 4$, $x_1 + 4x_2 - 8x_3 = 7$, $-3x_1 - 7x_2 + 9x_3 = -6$.
- b) Solve the following system of equations by Gauss Seidel iteration method. Carry out three iterations, taking $(0, 0, 0)$ as the initial approximation:
 $2x + y + 6z = 9$, $8x + 3y + 2z = 13$, $x + 5y + z = 7$.

2. a) By LU decomposition method, factorize the matrix A as LU . Given
- $$A = \begin{bmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{bmatrix}.$$
- b) For what values of a and b , the system has (i) no solution (ii) unique solution (iii) infinite number of solutions:
 $x + 2y + 3z = 6$, $x + 3y + 5z = 9$, $2x + 5y + az = b$.
3. a) If $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$. Verify that (i) $\text{tr}(B^{-1}AB) = \text{tr}(A)$ (ii) $\text{tr}(A) = \text{sum of the eigen values of } A$.
- b) Find all eigen values of $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$. Find an eigen vector corresponding to the largest eigen value.
4. a) Determine the largest eigen value and the corresponding eigen vector of $A\mathbf{x} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \mathbf{x}$ by power method. Take $X = (1 \ 1 \ 1)^T$. Carry out four iterations.
- b) Diagonalize $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ if possible.

21CS401 MSE-II — July 2023

1. a) If G and W are subspaces of a vector space V over the field F then prove that $G \cap W$ is also a subspace of V .
- b) Find a basis for (i) Column Space (ii) Null Space of $A = \begin{bmatrix} 1 & 4 & 5 & 9 \\ 1 & 6 & 5 & 1 \\ 3 & 4 & 8 & -3 \end{bmatrix}$.
2. a) (i) Define linearly independent vectors.
(ii) Check whether the set of vectors $\{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$ is linearly independent or linearly dependent.
- b) Let W be the subspace of R^4 spanned by $\{(1, 1, 1, 1), (1, 2, 3, 2), (2, 5, 6, 4), (2, 6, 8, 5)\}$.
(i) Find a basis for W .
(ii) Extend the basis of W to a basis of R^4 .
3. a) Check whether a transformation $T : R^2 \rightarrow R^2$ defined by $T(x, y) = (3x + 2y, 3x - 4y)$ is a linear transformation or not.
- b) (i) State Rank-Nullity theorem.
(ii) The linear transformation $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Find the nullity of T and rank of T .
4. a) (i) Define null space of a linear transformation $T : V \rightarrow W$.
(ii) Prove that null space of T is a subspace of V .
- b) A linear transformation defined by $T(x, y, z) = (6x + 2y - 4z, x - 5y + 3z)$. Find the matrix of linear transformation relative to the bases $B_1 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $B_2 = \{(1, 3), (2, 5)\}$.

21CS401 MSE-I — June 2023

1. a) Find the general solution of the system whose augmented matrix is $\begin{bmatrix} 0 & -7 & 0 & 6 & 5 \\ 0 & -1 & 7 & -4 & 9 \\ 0 & 0 & -2 & 2 & 0 \\ 1 & 2 & -3 & 4 & 0 \end{bmatrix}$.
- b) Find LU factorization of $A = \begin{bmatrix} 1 & 3 & 5 & 14 \\ -1 & -5 & 8 & 4 \\ 4 & 9 & 2 & -5 \\ -2 & -4 & 7 & 5 \end{bmatrix}$.
2. a) (i) Define basic and free variables.
(ii) Which are the basic variables in the system $AX = B$ where $A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix}$ and $B = \begin{bmatrix} 7 \\ -1 \\ 4 \end{bmatrix}$?
(iii) Determine the rank of the coefficient matrix.
b) Solve the following system of equations by inverse method: $x+2y+z = 4$, $2x+3y-z = 3$, $3x+y+2z = 3$.
3. a) (i) Define trace and spectrum of a square matrix A .
(ii) If $A = \begin{bmatrix} 0 & 1 \\ 1 & 2 & 3 \\ 3 & 9 & 1 \end{bmatrix}$, find the trace of AA^T .
b) Diagonalize $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ if possible.
4. a) Determine the eigen values and eigen vectors of $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$.
b) Using Cayley Hamilton Theorem find the inverse of $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$.

MA2006-1 MSE-I — February 2024

1. a) Define the rank of a matrix. Reduce the following matrix to row echelon form and hence find its rank:
 $\begin{bmatrix} 2 & -4 & 3 & -1 & 0 \\ 1 & -2 & -1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$.
b) Define the consistency of linear system of equations. Test whether the following equations are consistent or not:
 $2x - 3y + 7z = 5$, $3x + y - 3z = 13$, $2x + 19y - 47z = 32$.
2. a) Solve the given system of equations by Gauss elimination method: $x + y + z = 6$, $x - y + 2z = 5$, $3x + y + z = 8$.

- b) Use Gauss-Seidel iteration method to solve: $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$. Start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry out three iterations.
3. a) Find the eigen vectors of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.
- b) Use Rayleigh's power method to find the largest eigen value and eigen vector of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ with the initial eigen vector as $[1 \ 1 \ 1]^T$. Carry out 4 iterations.
4. a) Define Eigen values and Eigen vector. Find the eigen values of the matrix (text unclear in OCR).
- b) Find the modal matrix and spectral matrix of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

19MA8X02 SEE — January 2023

1. a) Test for consistency and solve the system of equations $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 4y + 7z = 30$.
- b) Find the rank of matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$ by reducing it to row echelon form.
2. a) Solve the system of equations by Gauss-Jordan method: $2x + y + 4z = 16$, $3x + 2y + z = 10$, $x + 3y + 3z = 16$.
- b) Solve by LU factorization method: $x + y + z = 9$, $2x - 3y + 4z = 13$, $3x + 4y + 5z = 40$.
3. a) Solve the system of equations by Gauss-Seidel method: $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$.
- b) Using Cayley-Hamilton theorem, find A^{-1} and A^4 given $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$.
4. a) Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.
- b) Diagonalize the matrix $A = \begin{bmatrix} 4 & -1 & 1 \\ -1 & 4 & -1 \\ 1 & -1 & 4 \end{bmatrix}$.
5. a) Reduce the quadratic form $3x^2 + 3y^2 + 3z^2 - 2xy - 2xz - 2yz$ to canonical form. What is its nature?
- b) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$.
6. a) Use Rayleigh's power method to find the largest eigenvalue and eigenvector of $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. Take $X_0 = [1 \ 0 \ 0]^T$. Carry out 5 iterations.

20IS401 Supplementary — September 2022

1.
 - a) Find the rank of matrix A by reducing it to row echelon form.
 - b) Test for consistency and solve the system of equations.
2.
 - a) Solve the system by Gauss-Jordan method.
 - b) Solve by LU factorization method.
3.
 - a) Find eigenvalues and eigenvectors of the given matrix.
 - b) Diagonalize the given matrix.
4.
 - a) Reduce the quadratic form to canonical form.
 - b) Verify Cayley-Hamilton theorem.
5.
 - a) Use Gauss-Seidel method to solve the system of equations.
 - b) Find the largest eigenvalue using Rayleigh's power method.

Microprocessor and Embedded Systems

CS3005-1 SEE — May 2025

1.
 - a) Write an ALP to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm.
 - b) Describe the roles of various status flags and control flags in Intel 8086 Microprocessor with the register format.
2.
 - a) Identify the addressing modes of the following instructions and justify the answer:
 - (i) MOV WORD PTR [SI], 20H
 - (ii) MOV ES:[1000H], 10H
 - (iii) MOV CX, NUM[BX+DI]
 - b) How various Shift Instructions work? Differentiate between arithmetic and logical shift instructions. Explain with examples.
3.
 - a) Develop an ALP to reverse a given string and check whether it is a palindrome or not.
 - b) Explain the flag register along with register format with respect to Intel 8086.
4.
 - a) Develop an ALP with procedures to perform the following:
 - (i) Read a string from the keyboard terminated by carriage return
 - (ii) Display the string
 - (iii) Display the length of the string
 - b) Explain the use of instruction queue in 8086 architecture.
5.
 - a) Write a Program to perform following operations: (i) Create a file (ii) Write "GOOD MORNING" in created file.
 - b) Develop an embedded C program to perform up-down counter on a 7-segment display using switches.
 - c) Describe the program status register of ARM processor with neat diagram.
6.
 - a) Write an ALP to do the following BIOS and DOS operations: Read a character from the KeyBoard. Add 4 lines and 4 columns to the current cursor position. Clear the entire screen and bring the cursor to newly calculated position and display the ASCII value of pressed character.
 - b) With code snippet, explain digitalWrite(), and analogRead() functions.
 - c) Explain the ARM core model and its functional units with a neat diagram.
7.
 - a) Give the coding template for the MOV instruction of the 8086 Intel microprocessor. Generate machine code for the following instruction: MOV CL, [1101].
 - b) Explain the Type 3 and Type 4 interrupts in 8086.
 - c) Differentiate between Minimum mode and maximum mode operation of 8086 microprocessor.
8.
 - a) Generate machine code for the following instruction of Intel 8086: MOV CX, 4012H[DI+BX].
 - b) What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling.
 - c) List the signals used in both minimum and maximum mode operation and explain any four signals in detail.

CS3005-1 SEE — May 2024

1.
 - a) Develop an ALP to search a key element in a list of 'n' 8-bit numbers using the Linear search algorithm. Display the position of the element if found, otherwise display the element not found.
 - b) How various Shift Instructions work? Differentiate between arithmetic and logical shift instructions. Explain with examples.
2.
 - a) Develop an ALP with procedures to perform the following: (i) Read a string from the keyboard terminated by carriage return (ii) Display the string (iii) Display the length of the string.
 - b) Explain the flag register along with register format with respect to Intel 8086.
3.
 - a) Develop an ALP to reverse a given string and check whether it is a palindrome or not.
 - b) List and explain the various addressing modes available in Intel 8086.
4.
 - a) Explain the following 8086 Assembler Directives with an example for each: (i) ASSUME (ii) SEGMENT (iii) INCLUDE (iv) DB.
 - b) Describe the program status register of ARM processor with neat diagram.
5.
 - a) Develop an ALP to read a string from the keyboard till carriage return and write into a file.
 - b) Explain the ARM core model and its functional units with a neat diagram.
6.
 - a) Develop an ALP procedure to convert two-digit Hexadecimal to equivalent ASCII values.
 - b) Bring out the various differences between RISC and CISC machines.
 - c) Develop an Embedded C program to read the input from 4x4 keypad and simulate operations as a calculator.
7.
 - a) Explain the following pin functions of 8086: (i) NMI (ii) READY.
 - b) Explain the Type 0 and Type 1 interrupts in 8086.
 - c) Give the template for the MOV instruction of 8086 microprocessor. Generate machine code for: (i) MOV CL, [BX] (ii) MOV 43H[SI], DH.
8.
 - a) Differentiate between Maximum mode and minimum mode operation of 8086 microprocessor.
 - b) What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling.
 - c) Explain the following maximum mode pin functions of 8086: (i) QS0, QS1 (ii) LOCK (iii) Request/Grant.

CS3005-1 SEE — July 2024

1.
 - a) Write an ALP to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm.
 - b) What is the difference between the following instructions? (i) JL UP and JB UP (ii) XOR AX, AX and MOV AX, 0.

- c) State the various assembler directives used and their meaning in the assembly language segment given below:
OUR_DATA SEGMENT
NUM1 DW 1234H
NUM2 DB 12H,34H,3DUP(0)
OUR_DATA ENDS
2. a) How various Shift Instructions work? Differentiate between arithmetic and logical shift instructions. Explain with examples.
b) Write an ALP to exchange two register contents AL and BL without using XCHG instruction and without using temporary register/memory location.
c) Explain the flag register along with register format with respect to Intel 8086.
3. a) Identify the addressing modes of following instructions and justify the answer: (i) MOV WORD PTR [SI], 20H (ii) MOV ES:[1000H], 10H (iii) MOV CX, NUM[BX+DI].
b) Explain the use of instruction queue in 8086 architecture.
c) Write an ALP to reverse a given string and check whether it is a palindrome or not.
4. a) Write a Program to perform following operations: (i) Create a file (ii) Write “GOOD MORNING” in created file.
b) Develop an embedded C program to perform up-down counter on a 7-segment display using switches.
c) Describe the program status register of ARM processor with neat diagram.
5. a) Write an ALP to do the following BIOS and DOS operations: Read a character from the KeyBoard. Add 4 lines and 4 columns to the current cursor position. Clear the entire screen and bring the cursor to newly calculated position and display the ASCII value of pressed character.
b) With code snippet, explain digitalWrite(), and analogRead() functions.
c) Explain the ARM core model and its functional units with a neat diagram.
6. a) Write an ALP to convert two ASCII values to the equivalent hexadecimal.
b) What is recursion? Write an ALP to find factorial of a number stored in data segment using recursion.
c) Write an Embedded C program to read the input from 4x4 keypad and simulate operations as a calculator.
7. a) Give the coding template for the MOV instruction of the 8086 Intel microprocessor. Generate machine code for MOV CL, [1101].
b) Explain the Type 3 and Type 4 interrupts in 8086.
c) Differentiate between Minimum mode and maximum mode operation of 8086 microprocessor.
8. a) Generate machine code for the following instruction: MOV CX, 4012H[DI+BX].
b) What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling.
c) List the signals used in both minimum and maximum mode operation and explain any four signals in detail.

CS3005-1 MSE-I — February 2024

1.
 - a) Describe the roles various status flags and Control flags in Intel 8086 Microprocessor. If the two numbers 1234H and 05A5H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer.
 - b) Assume that the address of 1st element in the following array is 2742H:
ARR DB 77H, 41H, 27H, 24H
What is the contents of DH and DL after the execution of the following instructions:
(i) LEA DX, ARR (ii) MOV DX, ARR. Justify your answers.
2.
 - a) Suppose that DS=1200H, BX=0100H, SI=0250H, DI=2024H, SS=1400H, BP=1500H, Starting address of ARRAY=0012H. Determine the physical address of the data:
(i) MOV AL, ARRAY[BX]
(ii) MOV AL, [BP+SI+200]
(iii) MOV DL, ARRAY[BX+DI]
 - b) Explain the following addressing Modes with example: (i) Direct Addressing Mode
(ii) Based Addressing Mode.
3.
 - a) Write an Assembly language program to find minimum element in a byte array.
 - b) Explain the following assembler directives with example: (i) ASSUME (ii) MACRO.
4.
 - a) Explain the following assembler instructions by giving an example for each: (i) DIV
(ii) DAA.
 - b) Write an Assembly Language Program to calculate LCM of two byte numbers.

21CS403 MSE-I — June 2023

1.
 - a) Explain general-purpose register and flag register of 8086 architecture.
 - b) Describe the difference between the following instructions:
(i) MOV AX, 2437H and MOV AX, [2437H]
(ii) MOV AL, [24374H] and MOV AX, [2436H]
Assume that [2436H]=23H, [2437H]=5EH and [2438H]=83H.
2.
 - a) Explain the working of following 8086 instructions along with example for each: (i) IMUL (ii) DAA (iii) LEA.
 - b) Indicate the addressing modes of the destination operand and calculate the physical address for the following instructions:
(i) ADD [DI], AL
(ii) AND [BX+DI], CL
Assume CS=1000H, DS=2000H, SS=3000H, ES=4000H, BX=0FFFH, DI=1A00H.
3.
 - a) Write an ALP program to read a character from keyboard. Display message “MICROPROCESSOR” if the key pressed is ‘M’ otherwise, do not display any message.
 - b) Write the 8086 instruction which will perform the following:
(i) Copy AX to a memory location at offset 245AH in the data segment.
(ii) Rotate the most significant bit of AL into the least significant bit position.
(iii) Mask the lower four bits of BL.
(iv) Set most significant bit of AX to a 1 but do not affect the other bits.
4.
 - a) Write an ALP to find maximum value in a byte array and display the maximum value.
 - b) Define Instruction Queue and explain the use of instruction queue in 8086 architecture.

21CS403 SEE — August 2023

1.
 - a) Briefly Explain General purpose registers and flag register with respect to Intel 8086.
 - b) Explain the working of following 8086 instructions along with example for each: (i) AAM (ii) DAA.
 - c) Assume that BX is 1020H and the content of 1020H=3FH; Content of 1021H=6EH. What is the content of CX register after execution of the statement MOV CX, [BX]? Justify your answer.
 - d) Write an ALP to calculate LCM of two byte numbers.
2.
 - a) What is the role of BIU and EU of Intel 8086 microprocessor? Explain BIU in detail.
 - b) Explain the following addressing Modes with instruction example: (i) Direct Addressing Mode (ii) Register Relative Addressing mode (iii) Based Indexed Addressing Mode.
 - c) Write an ALP to reverse a given string and check whether it is a palindrome or not.
3.
 - a) How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example.
 - b) Briefly explain the following Assembler directives used in Intel 8086 microprocessor: (i) DD (ii) SEGMENT (iii) EQU (iv) ASSUME.
 - c) Assume that the address of 1st element in the following array is 2742H:
ARR DB 77H, 41H, 27H, 24H
What is the contents of DX after the execution of following instructions: (i) LEA DX, ARR (ii) MOV DX, ARR. Justify your answers.
4.
 - a) Write an ALP to display “FIREHELP” using 7-segment display.
 - b) Write an embedded program to left scrolling effect on 7-segment display.
 - c) Describe Embedded system and its components.

19EC503 Supplementary — September 2022

1.
 - a) Explain the flag register format of 8086 microprocessor. With an example verify the status of flag register.
 - b) Describe the BIU block of 8086 microprocessor with a neat diagram.
2.
 - a) Explain the following addressing modes with an example for each: (i) Register addressing mode (ii) Immediate addressing mode (iii) Direct addressing mode.
 - b) Explain shift and rotate instructions of 8086.
3.
 - a) Write an ALP to find the largest number in a given array.
 - b) Write an ALP to arrange the given data bytes in ascending order.
4.
 - a) Explain the following assembler directives: (i) ASSUME (ii) SEGMENT (iii) DB (iv) DW.
 - b) Write an ALP to find factorial of a number using recursive procedure.
5.
 - a) Explain Type 0 and Type 1 interrupts in 8086.
 - b) Explain the minimum mode pin configuration of 8086.

20EC503 SEE — January 2023

1.
 - a) Draw and explain the internal architecture of 8086 microprocessor.
 - b) Explain various addressing modes of 8086 with an example for each.
2.
 - a) Explain the flag register of 8086 microprocessor. Find the status of the flags after the addition of 2B6AH and 54C8H.
 - b) Write an ALP to find the smallest number in a given array of 16-bit numbers.
3.
 - a) Write an ALP to sort numbers in ascending order using bubble sort.
 - b) Write an ALP to multiply two 16-bit numbers.
4.
 - a) Explain the following assembler directives with example: (i) ASSUME (ii) SEGMENT (iii) PROC (iv) DB (v) MACRO.
 - b) Write an ALP using string instructions to reverse a string.
5.
 - a) Explain minimum and maximum mode of operation of 8086.
 - b) What is interrupt? Explain types of interrupts in 8086 microprocessor.

20IS501 SEE — January 2023

1.
 - a) Draw and explain the internal architecture of 8086 microprocessor.
 - b) Explain various addressing modes of 8086 with an example for each.
2.
 - a) Explain the flag register of 8086 microprocessor.
 - b) Write an ALP to find the largest element in a given array.
3.
 - a) Write an ALP to sort numbers in ascending order.
 - b) Explain the following instructions with examples: (i) DAA (ii) IMUL (iii) LEA.
4.
 - a) Explain the following assembler directives: (i) SEGMENT (ii) ASSUME (iii) DB (iv) DW (v) MACRO.
 - b) Write an ALP using procedures to read a string, display it, and count characters.
5.
 - a) Explain minimum mode and maximum mode of 8086.
 - b) What is an interrupt? Explain interrupt handling in 8086.

Computer Organization and Architecture

CS2101-1 SEE — July 2024

1.
 - a) What are the basic functional units of a computer? Explain the basic operational concepts of a computer with a neat diagram.
 - b) Discuss the bus structure of a computer.
2.
 - a) What is an instruction? Explain the various instruction formats with examples.
 - b) Discuss the various types of instructions used in a computer.
3.
 - a) What is addressing mode? Explain the various addressing modes with suitable examples.
 - b) Discuss subroutine calling conventions.
4.
 - a) Explain the basic concepts involved in I/O operations.
 - b) Write a note on interrupts and interrupt handling.
5.
 - a) Explain memory mapping and its types.
 - b) Describe the various types of ROM.
6.
 - a) What is cache memory? Explain cache memory mapping techniques.
 - b) Explain the concept of virtual memory.
7.
 - a) Explain pipelining with a diagram. What are pipeline hazards?
 - b) Discuss the various hazard resolution techniques.
8.
 - a) Explain the design of a hardwired control unit.
 - b) What is microprogrammed control? Explain with a diagram.

Innovation and Design Thinking

ME1654-1 SEE — May 2025

1.
 - a) What is design thinking? Explain with an example.
 - b) Explain 9 qualities of a good design.
2.
 - a) What are the characteristics of a good design? Explain with an example.
 - b) Empathetically collect data from a regular college going person looking for a shoe for daily use using 360° approach which can be also used for jogging and playing football.
3.
 - a) With a neat sketch show agile model of project management.
 - b) With the help of a flow chart explain the process modelling in design thinking.
4.
 - a) With a neat sketch show waterfall model.
 - b) What are the advantages of business process model in design thinking?
5.
 - a) Johnson is a bank manager, and he is looking for a walk in wardrobe design for his newly built house. Formals, casuals, semi formals, daily use clothes and other related accessories compartments are must. He has a son who is currently enrolled in college and a daughter who is 4 years old and goes to preschool. His wife maintains a successful career in her corporate field. His mother is a homemaker who has a strong affinity for gold and silver ornaments. His father is an ex-service man retired as a major. Ideate a walk in wardrobe for this family located in one common space so that all of their clothing and other necessary requirements are taken care in one convenient location. Take into account all essential inputs.
6. There is a 1 acre farm land which is equivalent to 22 numbers of 30X40 Sq ft. site. This land is very close to forest and gets all the serenity and tranquility of nearby forest with some wild animals habitat like forest giant squirrel, monkeys, deer, wild boar, snakes and some varieties of native birds. The land is placed in a downhill making the rain water flow downwards and pass through the channel located at the center of the land. A corporate working engineer is looking to build a farm house on this land. He has plans of having medicinal plants in one section of the land, some fruit garden in the other section, a small lake to cultivate some fishes in the third section which can in turn serve as the source of water for the land, fourth and fifth sections for dog kennels having 3 different breeds and a cow shed holding around 5 cows, respectively. Empathize with the engineer, define his requirements and ideate the farm land layout for his ease of operation and maintenance with necessary assumptions. He is ready to invest not more than 20 lakhs on this land. Also show the respective price allotment to each section.

ME1654-1 MSE-I — February 2025

1.
 - a) Explain the following with the help of a common example: (i) Creativity (ii) Invention (iii) Innovation.
 - b) Explain with sketch agile model of project management.
2.
 - a) Define design thinking process. Write the characteristics of design thinking process.
 - b) Explain with sketch the design thinking in business process modelling.
3. How can we empathetically gather insights from an individual seeking a water bottle that supports both their fitness and health goals while also being ideal for trekking application using 360-degree design thinking approach.

4. How can we empathetically collect insights from a business professional seeking a wallet or a purse that securely holds credit/debit cards while addressing the fear of purse or wallet being stolen along with data security of credit/debit cards, using 360-degree design thinking approach.

Supplementary and Older Exam Papers

19CS502 DBMS Supplementary — September 2022

1.
 - a) What are the advantages of DBMS over file system? Explain.
 - b) With a neat diagram, explain the three-schema architecture of DBMS.
2.
 - a) Explain the various types of attributes with examples.
 - b) Design an ER diagram for a Banking system. State the assumptions made.
3.
 - a) Explain the ER to relational mapping algorithm.
 - b) What are the relational algebra operations? Explain UNION, INTERSECTION and DIFFERENCE operations.
4.
 - a) Write SQL queries for the following using Employee, Department, and Project tables.
 - b) What is a view? Explain with an example.
5.
 - a) Explain 1NF, 2NF and 3NF with examples.
 - b) What are functional dependencies? Explain with examples.

20IS402 DAA Supplementary — September 2022

1.
 - a) Explain the properties of an algorithm with an example.
 - b) Write the algorithm for selection sort and analyze its time complexity.
2.
 - a) Define Big-O, Big-Omega and Big-Theta notations.
 - b) Write the merge sort algorithm and trace it for the list: 38, 27, 43, 3, 9, 82, 10.
3.
 - a) Write the brute force string matching algorithm and analyze its complexity.
 - b) Apply quick sort to sort the array: 5, 3, 8, 4, 2, 7, 1.
4.
 - a) What is Divide and Conquer? Explain binary search algorithm.
 - b) Write the algorithm for bubble sort and analyze its complexity.

NMAM Institute of Technology, Nitte

Off-Campus Centre of Nitte (Deemed to be University)

COMPREHENSIVE EXAMINATION QUESTION BANK

IV Semester B.E./B.Tech (CBCS)

Academic Years: 2022–2025

MSE-I, MSE-II, SEE & Supplementary Examinations

Subjects Included:

Part A:	MCQs + Descriptive (From Images)
MA2005-1	Linear Algebra and Numerical Methods
CS2102-1	Database Management Systems
CS3004-1	Design and Analysis of Algorithms
ME1654-1	Innovation and Design Thinking
21CS403	Microprocessor and Microcontrollers
21CS404	Software Engineering & Project Management
Part B:	Descriptive Only (From PDF Source)
CS2103-1/AD1103-1	Software Engineering & Project Management
CS3005-1	Microprocessor and Embedded Systems
CS2101-1	Computer Organization and Architecture

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Part I**MCQs + Descriptive Questions (From Images)****1 Linear Algebra and Numerical Methods (MA2005-1)****1.1 MSE-I – February 2025****NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025****MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS**

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 15

Part – A: Multiple Choice Questions ($3 \times 1 = 3$ marks)

- For what values of k will the augmented matrix $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 2 & 1 & 3 \\ 1 & 2 & k & k+1 \end{array} \right]$ correspond to an inconsistent system of linear equations?
(A) $k \neq 1$ (B) $k = 0$ (C) $k = 1$ (D) All values of k
- If $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$, find $(AB)^{-1}$.
(A) $\begin{bmatrix} 1 & 1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & -1/2 \\ -1/6 & 1/2 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -1/3 \\ -1/6 & 1/2 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & -1/3 \\ 1/6 & 1/2 \end{bmatrix}$
- Given a matrix A , column 1 and column 3 are pivot columns, then the dimension of Col A is:
(A) 1 (B) 2 (C) 3 (D) 4

Part – B: Descriptive Answer Questions ($2 \times 6 = 12$ marks)**Unit – I**

- a) Find the eigenvalues of $A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$. Also determine the eigenvectors corresponding to each eigenvalue. [6 marks]
b) Reduce the matrix $A = \begin{bmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{bmatrix}$ to the form $PA = LU$, using partial pivoting. [6 marks]

Unit – II

- a) Solve by Gauss-Seidel method: $10x + y + z = 12$; $2x + 10y + z = 13$; $x + y + 5z = 7$. (3 iterations) [6 marks]
b) Find the dominant eigenvalue and eigenvector of $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ using Rayleigh's Power method (4 iterations). [6 marks]

1.2 MSE-II – April 2025**NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025****MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS**

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions ($4 \times 1 = 4$ marks)

- Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, x_1)$. Then $T^{-1}(x_1, x_2)$ equals:
(A) $(x_2, x_1 - x_2)$ (B) $(x_2, x_1 + x_2)$ (C) $(x_1, x_1 - x_2)$ (D) $(x_1, x_1 + x_2)$
- The set $\{1, x, x^2\}$ forms a basis for:
(A) P_1 (B) P_2 (C) P_3 (D) P_4
- Which is a subspace of the given vector space?
(A) Set of all polynomials of degree exactly 3, in P_3
(B) Set of all polynomials with positive coefficients, in P
(C) Set of all 2×2 matrices with positive entries
(D) Set of all symmetric 2×2 matrices, in $M_{2 \times 2}$
- Given matrix $A_{4 \times 5}$ with $\text{rank}(A) = 3$, then $\dim(\text{Null}(A))$ is:
(A) 1 (B) 2 (C) 3 (D) 4

Part – B: Descriptive ($2 \times 8 = 16$ marks)**Unit – I**

- a) Let $T : P_2 \rightarrow P_2$ be the transformation $p(t) \mapsto p(t + 2)$. Find image of $p(t) = 3 - 2t + t^2$. Show T is linear. Find matrix with basis $B = \{1, t, t^2\}$. [8 marks]
b) Let $\vec{v}_1 = (1, -2, 3)$, $\vec{v}_2 = (5, 6, -1)$, $\vec{v}_3 = (3, 2, 1)$. Show $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is a basis for $\mathbb{R}^3$. [8 marks]

Unit – II

- a) Use Gram-Schmidt to construct orthonormal basis from $B = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$. Find QR factorization. [8 marks]
b) Use coordinate vectors to test linear independence of $S = \{1 - 2t^2 - t^3, t + 2t^3, 1 + t - 2t^2\}$. [8 marks]

1.3 SEE – May 2025**NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***Fourth Semester B.Tech (CBCS) Degree Examinations, May 2025****MA2005-1 – LINEAR ALGEBRA AND NUMERICAL METHODS**

(For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours

Max. Marks: 100

Part – A: MCQs (20 × 1 = 20 marks)

- For what value of k , the augmented matrix corresponds to inconsistent system?

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 3 \\ 3 & k & -6 & 9 \\ 1 & -3 & 2 & 5 \end{array} \right]$$
 (A) $k = 1$ (B) $k = 9$ (C) $k = -3$ (D) None
- If A and B are matrices such that AB exists and is square, then:
 (A) Both are square of same order (B) Number of rows of A = columns of B
 (C) BA need not exist (D) One of them is identity
- Columns 1, 2, 5 are pivot columns. Which is NOT necessarily true?
 (A) $\dim(\text{Col } A) = 3$ (B) $\dim(\text{Row } A) = 3$ (C) $\text{rank}(A) = 3$ (D) $\dim(\text{Null } A) = 3$
- If λ is eigenvalue of A , eigenvalue of A^{-1} is: (A) λ (B) $-\lambda$ (C) λ^2 (D) $1/\lambda$
- If $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ then $A^3 - 3A^2 - A + 3I =$ (A) $A + I$ (B) I (C) $2A$ (D) 0
- Eigenvalues of $P^{-1}AP$ are same as: (A) PA (B) A (C) P (D) Cannot say
- Gauss-Seidel: arrange so that diagonal elements are: (A) Any order (B) Zero (C) Larger than others (D) Cannot solve
- Which is a direct method? (A) Jacobi (B) Gauss-Seidel (C) LU Decomposition (D) Power method
- If λ is eigenvalue of A , then $\lambda^2 + 1$ is eigenvalue of: (A) $A^2 + I$ (B) $A^2 + A$ (C) $2A$ (D) $(A + I)^2$
- If $1 + x, 1 + x + x^2, ax^2 + x^3$ are linearly dependent, $a =$ (A) 1 (B) 2 (C) 3 (D) 4
- Dimension of $W = \{(a, b, c, d) : a - 2b + c - d = 0\}$ is: (A) 1 (B) 2 (C) 3 (D) 4
- $V =$ set of 2×2 symmetric matrices, dimension is: (A) 1 (B) 2 (C) 3 (D) 4
- For $T : V \rightarrow W$, $T^{-1}(U)$ is subspace of: (A) V (B) W (C) Both (D) Neither
- $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be matrix multiplication: (A) Only if $m = n$ (B) Only if $m \neq n$ (C) Always (D) Never
- If $\vec{u}, \vec{v}$ orthogonal: $\|\vec{u} + \vec{v}\|^2 =$ (A) $\|\vec{u}\|^2 + \|\vec{v}\|^2$ (B) $\|\vec{u}\|^2 - \|\vec{v}\|^2$ (C) Product (D) Quotient
- $\vec{u} = (1, 2, -3)$, $\vec{v} = (a, 1, 1)$ orthogonal if $a =$ (A) 1 (B) 2 (C) 3 (D) 5
- Linear combination of rows of A : (A) $\vec{x}^T A$ (B) $A\vec{x}$ (C) $\vec{x}A^T$ (D) $A^T\vec{x}$
- Angle between $\vec{u} = (1, 2, -1, 0)$ and $\vec{v} = (-1, 1, 1, 2)$: (A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) $2\pi/3$
- Which defines inner product on $\mathbb{R}^2$? (A) $u_1v_1 - u_2v_2$ (B) $u_1v_1 + 2u_2v_2$ (C) $u_1^2v_1^2 + u_2^2v_2^2$ (D) $u_1v_2 - u_2v_1$
- For orthonormal basis: $\|\vec{x}\|^2 =$ (A) $\sum |\langle \vec{x}, \vec{u}_i \rangle|^2$ (B) $\sum \langle \vec{x}, \vec{u}_i \rangle$ (C) $\sum |\langle \vec{x}, \vec{u}_i \rangle|$ (D) None

Part – B: Descriptive (80 marks)

[Questions 1-6 with eigenvalues, LU factorization, Gauss-Seidel, Power method, vector spaces, linear transformations, Gram-Schmidt, QR factorization, SVD, least squares - each 16 marks]

2 Database Management Systems (CS2102-1)

2.1 MSE-I – February 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 15

Part – A: MCQs ($3 \times 1 = 3$ marks)

1. A ____ is a unit of storage containing records: (A) Block (B) Field (C) Tuple (D) File
2. NOT a component of ER Model: (A) Entity (B) Attribute (C) Table (D) Relationship
3. Used to uniquely identify a record: (A) Foreign Key (B) Candidate Key (C) Primary Key (D) Super Key

Part – B: Descriptive (12 marks)

1. a) Draw ER diagram for University Database with Student, Course, Instructor, Department. [6 marks]
b) Explain three-schema architecture of DBMS with diagram. [6 marks]
2. a) Movie database relational algebra queries. [6 marks]
b) Explain different types of keys in DBMS with examples. [6 marks]

2.2 MSE-II – April 2025

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 20

Part – A: MCQs ($4 \times 1 = 4$ marks)

1. “No component of Primary Key accepts Null” is: (A) Referential Integrity (B) Domain Integrity (C) Entity Integrity (D) Tuple Integrity
2. Add new column to table: (A) ALTER (B) UPDATE (C) INSERT (D) MODIFY
3. BCNF: every non-trivial FD $X \rightarrow Y$: (A) X is super key (B) Y is prime (C) X is candidate key (D) Y is super key
4. Combine tuples based on condition: (A) Union (B) Join (C) Projection (D) Selection

Part – B: Descriptive (16 marks)**Suppliers-Parts-Catalog SQL Queries:**

Suppliers(sid, sname, address)

Parts(pid, pname, color)

Catalog(sid, pid, cost)

- i) Find names of suppliers who supply some red part.
- ii) Find sids of suppliers who supply some red or green part.
- iii) Find sids who supply red part or are at 221 Packer street.
- iv) Find pids supplied by at least two different suppliers.

Functional Dependencies: $R(A, B, C, D, E)$ with FDs: $\{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. Find candidate keys. Check equivalence.

Normalization: $R(A, B, C, D)$ with $\{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$. Find candidate keys. Check 3NF. Decompose if needed.

2.3 SEE – May 2025**NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***Fourth Semester B.Tech (CBCS) Degree Examinations, May 2025****CS2102-1 – DATABASE MANAGEMENT SYSTEMS**

(For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours

Max. Marks: 100

Part – A: MCQs (20 marks)

[20 MCQs covering: DBMS characteristics, ER diagrams, weak entities, relational model, relational algebra, SQL clauses, DML commands, functional dependencies, normal forms, ACID properties, 2PL, timestamp ordering]

Part – B: Descriptive (80 marks)

[8 questions on ER diagrams, relational algebra, SQL queries, normalization, FD closure, lossless decomposition, transactions, concurrency control]

3 Design and Analysis of Algorithms (CS3004-1)**3.1 MSE-I – February 2025****NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025****CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS**

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 15

Part – A: MCQs (3 marks)

1. $T(n) = 7n^3 + 10n^2 + 4$ belongs to: (A) $O(n)$ (B) $O(n^2)$ (C) $O(n^3)$ (D) $O(n^4)$
2. Time complexity of $T(n) = 2T(n/2) + n$: (A) $O(n)$ (B) $O(n \log n)$ (C) $O(n^2)$ (D) $O(\log n)$
3. Minimum comparisons to find minimum in array of n elements: (A) n (B) $n - 1$ (C) $n + 1$ (D) $2n$

Part – B: Descriptive (12 marks)

1. a) Solve TSP using brute force for 4-vertex graph with edges: 1-2:10, 2-3:35, 3-4:25, 4-1:20, 1-3:15, 2-4:30. [6 marks]
b) Quick Sort pseudocode. Trace for [38, 27, 43, 3, 9, 82, 10]. [6 marks]
2. a) Merge Sort algorithm with time complexity derivation. [6 marks]
b) Tower of Hanoi: recursive algorithm and time complexity. [6 marks]

3.2 MSE-II – April 2025**NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025****CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS**

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 20

Part – A: MCQs (4 marks)

1. Topological sorting requires: (A) Tree (B) DAG (C) Complete graph (D) Bipartite graph
2. DFS uses: (A) Queue (B) Heap (C) Priority queue (D) Stack
3. Distribution counting sort complexity: (A) $\Theta(n^2)$ (B) $O(n \log n)$ (C) $O(2^n)$ (D) $\Theta(n)$
4. Dynamic programming builds from: (A) Overlapping subproblems (B) Single recursive call (C) Greedy (D) Divide and conquer

Part – B: Descriptive (16 marks)

1. a) Insertion sort: Sort E, X, A, M, P, L, E alphabetically. [4 marks]
b) Heap sort: Sort 90, 89, 56, 72, 67, 12, 21, 34, 45. [4 marks]
2. a) AVL tree construction for {2, 9, 3, 12, 15, 8, 11}. [4 marks]
b) Topological sorting using Source Removal method. [4 marks]
3. a) Comparison Counting sort algorithm. Compare with Selection and Bubble sort. [4 marks]
b) Horspool algorithm: Search TIME in “A STITCH IN TIME SAVES NINE”. [4 marks]

3.3 SEE – May 2025**NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***Fourth Semester B.Tech (CBCS) Degree Examinations, May 2025****CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS**

(For CS, IS, AM, CC, AD, CB)

Duration: 3 Hours

Max. Marks: 100

Part – A: MCQs (20 marks)

[20 MCQs on Quick sort, Binary search, Linear search, asymptotic analysis, recursion, Divide and Conquer, Dynamic programming, Floyd-Warshall, Horspool, Counting sort, Heap sort, AVL trees, B-trees, Branch and Bound, Backtracking, Dijkstra, Huffman codes]

Part – B: Descriptive (80 marks)

1. Performance Analysis, Selection sort [16 marks]
2. Non-recursive analysis, Merge sort [16 marks]
3. Heap construction, Horspool [16 marks]
4. B-Tree insertion, Topological sorting [16 marks]
5. Quick sort, Class P/NP/NP-Complete [16 marks]
6. Job Assignment (Branch and Bound), Subset Sum (Backtracking) [16 marks]

4 Innovation and Design Thinking (ME1654-1)**4.1 MSE-I – February 2025****NMAM INSTITUTE OF TECHNOLOGY, NITTE***Off-Campus Centre of Nitte (Deemed to be University)***IV SEM B.Tech (CBCS) Mid Semester Examinations - I, February 2025****ME1654-1 – INNOVATION AND DESIGN THINKING**

(For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 15

Part – A: MCQs (3 marks)

1. Design Thinking focuses on: (A) Strict rules (B) Understanding user needs (C) Minimizing collaboration (D) Avoiding experimentation
2. “Empathy” means: (A) Finding user faults (B) Understanding users’ feelings and motivations (C) Designing on assumptions (D) Testing without users
3. “Define” stage is about: (A) Generating ideas (B) Articulating problem from empathy insights (C) Building prototypes (D) Observing users

Part – B: Descriptive (12 marks)

1. a) Five stages of Design Thinking with examples. [6 marks]
b) Importance of empathy. Methods to build empathy. [6 marks]
2. a) Ideation phase. Two ideation techniques. [6 marks]
b) Prototyping: types of prototypes. [6 marks]

5 July 2023 & June 2022 MSE-II Papers**5.1 Linear Algebra (21CS401) – July 2023**

[MCQs on dimension of symmetric matrices, linear transformations, basis, null space + Descriptive on basis for P_2 , kernel/range, Gram-Schmidt, least squares]

5.2 DAA (20CS402) – July 2023

[MCQs on Heap Sort, AVL trees, DFS, Horspool + Descriptive on Heap Sort, AVL construction, DFS/BFS, Horspool string matching]

5.3 Microprocessor (21CS403) – July 2023

[MCQs on 8086 data bus, loop counter, physical address, flags + Descriptive on 8086 architecture, addressing modes, ALP for largest element, interrupt structure]

5.4 SEPM (21CS404) – July 2023

[MCQs on Waterfall model, SRS, non-functional requirements, Agile + Descriptive on Waterfall model, Agile vs Waterfall, SRS components, Use Case diagram]

5.5 Linear Algebra (20CS401) – June 2022

[MCQs on determinants, linear dependence, dimension, Rank-Nullity + Descriptive on subspaces, basis, linear transformations, orthonormal basis, projections]

5.6 DAA (21IS402) – June 2022

[MCQs on Quick Sort, B-tree, topological sorting, BFS + Descriptive on Quick Sort, B-tree construction, topological sorting, Horspool]

Part II

Descriptive Questions Only (From PDF Source)

6 Software Engineering & Project Management

6.1 CS2103-1 SEE – May 2025

1. a) Essential attributes of good software. Fundamental SE activities.
b) Incremental Development with diagram.
2. a) ACM/IEEE Code of Ethics.
b) Boehm's Spiral Model with diagram.
3. a) Structure of Requirements Document.
b) Types of Non-Functional Requirements with diagram.
4. a) Requirements Elicitation and Analysis Process with diagram.
b) Scenario for MHC-PMS medical history. Use cases.
5. a) System Testing.
b) Five key UML diagram types with purposes.
6. a) Concept of system models.
b) Language processing system architecture.
7. a) Four test strategies for software development.
b) ATM System software architecture.
8. a) Project scheduling concept and significance.
b) Configuration management concept.
9. a) Key stages in component measurement.
b) Risk identification in project management.

6.2 CS2103-1 SEE – May 2024

1. a) Software development life cycle.
b) Waterfall vs Spiral model.
2. a) V-model in plan-driven process.
b) Boehm's spiral model.
3. a) Functional vs non-functional requirements.
b) Incremental model advantages/disadvantages.
4. a) Data flow model (insulin pump example).
b) Acceptance testing process stages.
5. a) Sequence models and Use-case models.
b) Extreme programming and practices.
6. a) Weather Station object classes and diagrams.
b) Agile methods and principles.
7. a) Risk and Risk management with categories.
b) Project planning process.
8. a) Software review process phases.
b) Software product metrics and types.

6.3 AD1103-1 SEE – May 2024

[Similar questions on software attributes, Waterfall model, SDLC, Scrum, requirements document, Agile principles, System Testing, activity diagrams, object classes, testing strategies, system models, architectural views, risk identification, configuration management, project scheduling, quality management]

6.4 AD1103-1 SEE – May 2025

[Questions on requirements document, requirement validation, interview types, SDLC phases, software attributes, Waterfall model, Scrum framework, System Testing, UML diagrams, system models, language processing, test strategies, ATM architecture, project scheduling, configuration management, measurement process, risk identification]

6.5 19IS8X83 SEE – May-June 2023

[Questions on Software Engineering definition, incremental development, requirements specification, software attributes, Waterfall model, requirement validation, design process, Boehm's spiral, state diagrams, object-oriented design, sequence diagrams, agile principles, MVC pattern, transaction processing, generalization, scrum process, risk management, communication factors, software pricing, project planning workflow, software review phases, cohesive groups]

6.6 CS2103-1 MSE-I – February 2025

[Questions on software attributes, Incremental Development, ACM/IEEE Code of Ethics, Spiral Model, Requirements Document structure, Non-Functional Requirements, Elicitation and Analysis, MHC-PMS scenarios]

6.7 CS2103-1 MSE-I – February 2024

[Questions on ethical issues, software applications, spiral model, software attributes, SRS document, incremental model, Waterfall vs incremental, non-functional requirements, software engineering discipline, system requirements, MHC-PMS, elicitation and analysis]

6.8 21CS404 MSE-I – June 2023

[Questions on software attributes, non-functional requirements, Waterfall model, application types, professional responsibility, Boehm's spiral, requirements document structure, elicitation process]

7 Database Management Systems (Additional)

7.1 CS2102-1 SEE – July 2024

[Questions on database characteristics, relational algebra operations, ER diagrams for flight management, grouping in relational algebra, aggregate functions, three-schema architecture, Sailors-Boats-Reserves queries, Employee-Works-Company-Manages queries, views, FD decomposition, lossless join, SQL constraints, order processing queries, normalization, 2PL protocol, indexing, interleaved execution anomalies, B+ tree insertion]

7.2 20CS502 SEE – January 2023

[Questions on database characteristics, flight management ER diagram, database design phases, structural constraints, recursive relationships, relational algebra queries, division, natural join, aggregate functions, Employee-Project queries, ER-to-Relational mapping, grouping, banking database SQL, SQL constraints, student enrollment queries, views, FD equivalence, informal design guidelines, normalization, minimal cover, 2PL, NoSQL aggregate models, interleaved execution, CAP theorem]

7.3 CS2102-1 MSE-I – February 2024

[Questions on Three Schema Architecture, mappings, ER diagram for Engineering College, database characteristics, domain and referential integrity, entity attributes, ER-to-relational conversion, good database design, characteristics of relations]

8 Design and Analysis of Algorithms (Additional)

8.1 CS3004-1 MSE-I – February 2024

[Questions on algorithm properties with linear search example, function value changes (log, linear, quadratic, exponential, sqrt), algorithm analysis framework, asymptotic notations, recursive factorial analysis, brute force sorting, Merge sort with analysis, brute force string matching]

8.2 21CS402 MSE-I – June 2023

[Questions on algorithm design and analysis process flow, recursive algorithm for 2^n , asymptotic notations with proofs, sequential search complexity, brute force string matching, Merge sort with alphabetical sorting, Quick sort with analysis, exhaustive search for knapsack]

8.3 21CS402 Makeup SEE – September 2023

[Questions on distinct elements algorithm, brute force string matching, Merge sort with worst case, sequential search, selection sort complexity, Quick sort, Big Oh and Big Theta, binary digits algorithm, Divide and Conquer, Binary search]

9 Linear Algebra (Additional)

9.1 MA2005-1 MSE-I – February 2024

[Questions on Gauss-Seidel with 4 iterations, power method, LU factorization, Cramers rule, augmented matrix solutions, eigenvalues and eigenvectors, eigenvalue properties, iterative methods conditions, determinant evaluation, matrix multiplications, echelon form, row and column space basis]

9.2 21CS401 MSE-II – July 2023

[Questions on dimension of symmetric matrices, linear transformations (one-to-one, onto), basis properties, null space subspace, basis for P_2 , kernel and range, Gram-Schmidt orthonormalization, least squares solution]

9.3 21CS401 MSE-I – June 2023

[Questions on eigenvalues, echelon form, homogeneous systems, consistent systems, Gauss elimination, LU factorization, power method, matrix diagonalization]

9.4 MA2006-1 MSE-I – February 2024

[Questions on power method convergence, LU decomposition, Gauss-Seidel convergence, eigenvalue problems, matrix operations]

9.5 19MA8X02 SEE – January 2023

[Questions on determinants, matrix inverse, linear independence, eigenvalues, eigenvectors, Cayley-Hamilton theorem, diagonalization, Jordan form, inner product spaces, Gram-Schmidt, QR factorization]

9.6 20IS401 Supplementary – September 2022

[Questions on linear systems, echelon form, LU factorization, iterative methods, eigenvalues, power method, diagonalization, orthogonalization]

10 Microprocessor and Embedded Systems

10.1 CS3005-1 SEE – May 2025

[Questions on 8086 architecture, memory segmentation, addressing modes, instruction set, ALP for string operations, array processing, arithmetic operations, interrupt handling, 8255 PPI, timer/counter operations]

10.2 CS3005-1 SEE – May 2024

[Questions on microprocessor evolution, 8086 pin diagram, bus interface unit, execution unit, memory organization, stack operations, procedure calls, interrupts, DMA, serial communication]

10.3 CS3005-1 SEE – July 2024

[Questions on 8086 internal architecture, flag register, addressing modes, instruction formats, ALP programs, macros, procedures, interrupt service routines, peripheral interfacing]

10.4 CS3005-1 MSE-I – February 2024

[Questions on microprocessor basics, 8086 architecture, memory segmentation, instruction execution, addressing modes, data transfer instructions]

10.5 21CS403 MSE-I – June 2023

[Questions on 8086 data bus width, registers, physical address calculation, flags, architecture, addressing modes with examples]

10.6 21CS403 SEE – August 2023

[Questions on 8086 comprehensive coverage, ALPs, interrupt handling, I/O interfacing, timer applications]

10.7 19EC503 Supplementary – September 2022

[Questions on microprocessor fundamentals, 8086 programming, peripheral interfacing]

10.8 20EC503 SEE – January 2023

[Questions on 8086 architecture, instruction set, assembly language programming, I/O interfacing]

10.9 20IS501 SEE – January 2023

[Questions on microprocessor systems, 8086 programming, interrupt systems, peripheral devices]

11 Computer Organization and Architecture

11.1 CS2101-1 SEE – July 2024

[Questions on computer organization fundamentals, instruction set architecture, CPU design, memory hierarchy, cache organization, pipelining, I/O systems, parallel processing]

12 Innovation and Design Thinking (Additional)

12.1 ME1654-1 SEE – May 2025

[Questions on Design Thinking process, Empathize stage methods, Define stage techniques, Ideate stage tools, Prototype types, Test stage approaches, innovation frameworks, creativity techniques, user research methods, design challenges]

13 Supplementary Papers

13.1 19CS502 DBMS Supplementary – September 2022

[Questions on database concepts, ER modeling, relational algebra, SQL, normalization, transaction management]

13.2 20IS402 DAA Supplementary – September 2022

[Questions on algorithm analysis, sorting algorithms, searching algorithms, graph algorithms, dynamic programming, greedy algorithms]

— End of Comprehensive Question Bank —

*This document contains ALL examination papers from:
SEE (May 2025, May 2024, July 2024, January 2023, August 2023)
MSE-I (February 2025, February 2024, June 2023)
MSE-II (April 2025, July 2023)
Supplementary (September 2022)*

*Subjects: Linear Algebra, DBMS, DAA, SEPM, Design Thinking,
Microprocessor, Computer Organization*

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024
MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS
(FOR AM, CCE, CSE, ISE, RI)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Solve the following system of equations and obtain the general solution.
$x_1 + 3x_2 - 5x_3 = 4$
$x_1 + 4x_2 - 8x_3 = 7$
$-3x_1 - 7x_2 + 9x_3 = -6$ | 5 | L*2 | 3 | 1 |
| b) Solve the following system of equations by Gauss Seidel iteration method. Carry out three iterations, taking (0,0,0) as the initial approximation.
$2x + y + 6z = 9$
$8x + 3y + 2z = 13$
$x + 5y + z = 7$ | 5 | L2 | 3 | 1 |
| 2. a) By LU decomposition method, factorize the matrix A as LU. Given
$A = \begin{pmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{pmatrix}$ | 5 | L2 | 3 | 1 |
| b) For what values of a and b, the system has (i) no solution
(ii) unique solution (iii) infinite number of solutions
$x + 2y + 3z = 6$; $x + 3y + 5z = 9$; $2x + 5y + az = b$ | 5 | L3 | 3 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$
Verify that (i) $tr(B^{-1}AB) = tr(A)$
(ii) $tr(A) = \text{sum of the eigen values of } A$. | 5 | L2 | 4 | 1 |
| b) Find all eigen values of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$. Find an eigen vector corresponding to the largest eigen value. | 5 | L2 | 4 | 1 |
| 4. a) Determine the largest eigen value and the corresponding eigen vector of $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ by power method. Take $X^{(0)} = (1 \ 1 \ 1)^T$. Carry out four iterations. | 4 | L3 | 4 | 1 |
| b) Diagonalize $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ if possible. | 6 | L2 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024

CS3004-1- DESIGN AND ANALYSIS OF ALGORITHMS
(FOR AIDS, AIML, CCE, CSE, ISE)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Explain the properties of an algorithm taking linear search as example.	05	L*2	1	1
	b) For each of the following functions, indicate how much the function value will change if its argument is increased fourfold. i) $\log_2 n$ ii) n iii) n^2 iv) 2^n v) $\sqrt{n}$	05	L3	1	2
2.	a) Explain the general framework for Analysis of an algorithm.	05	L2	1	1
	b) Define 3 asymptotic notations and Determine whether the following assertions are true or false i) $10n^2+9 \in O(n^2)$ ii) $n(n+1)/2 \in \Theta(n^2)$	05	L3	1	2
		Unit – II			
3.	a) Write the general plan for analyzing the efficiency of recursive algorithm. Write the pseudo code to find factorial of a number and analyze its efficiency.	06	L3	1	2
	b) Give two brute force algorithms to sort the given list of integers.	04	L2	2	1
4.	a) Illustrate Merge sort algorithm and analyze its efficiency. Sort the following list using Merge sort. 6, 14, 8, 2, 15, 10, 7, 5	06	3	2	2
	b) Show the comparisons the Brute force string matching makes for the pattern $P=0001$ in the text $T=000010001010101$.	04	3	2	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024
CS2102-1 – DATABASE MANAGEMENT SYSTEMS
 (For AD, AM, CCE, CSE, ISE,)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any **One** full question from each Unit.

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Describe with a neat diagram of Three Schema Architecture. Why do we need mappings among schema levels?	4	L*2	1	1
	b) Design and construct an ER diagram for "Engineering College Management System. Indicate all keys, constraint and assumption that are made.	6	L3	2	2
2.	a) Discuss the main characteristics of database approach.	4	L2	1	1
	b) Explain how domain constraints and referential integrity constraints are specified in relational model.	6	L2	2	1

Unit – II

3.	a) Define an Entity attribute. Explain different types of attribute that occur in an E-R diagram model with an example.	5	L2	1	1
	b) Outline the steps to covert the basic E-R model to relational database scheme.	5	L2	2	1
4.	a) For a good database design relation schema should be such that matching attributes must be primary key or foreign key otherwise it leads to spurious tuples". Justify.	5	L3	2	2
	b) Discuss the characteristics of relations.	5	L2	2	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024
MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS
(FOR AM, CCE, CSE, ISE, RI)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) Solve the following system of equations and obtain the general solution.
$x_1 + 3x_2 - 5x_3 = 4$
$x_1 + 4x_2 - 8x_3 = 7$
$-3x_1 - 7x_2 + 9x_3 = -6$ | 5 | L*2 | 3 | 1 |
| b) Solve the following system of equations by Gauss Seidel iteration method. Carry out three iterations, taking (0,0,0) as the initial approximation.
$2x + y + 6z = 9$
$8x + 3y + 2z = 13$
$x + 5y + z = 7$ | 5 | L2 | 3 | 1 |
| 2. a) By LU decomposition method, factorize the matrix A as LU. Given
$A = \begin{pmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{pmatrix}$ | 5 | L2 | 3 | 1 |
| b) For what values of a and b, the system has (i) no solution (ii) unique solution (iii) infinite number of solutions
$x + 2y + 3z = 6$; $x + 3y + 5z = 9$; $2x + 5y + az = b$ | 5 | L3 | 3 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$
Verify that (i) $tr(B^{-1}AB) = tr(A)$
(ii) $tr(A) = \text{sum of the eigen values of } A$. | 5 | L2 | 4 | 1 |
| b) Find all eigen values of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$. Find an eigen vector corresponding to the largest eigen value. | 5 | L2 | 4 | 1 |
| 4. a) Determine the largest eigen value and the corresponding eigen vector of $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ by power method. Take $X^{(0)} = (1 \ 1 \ 1)^T$. Carry out four iterations. | 4 | L3 | 4 | 1 |
| b) Diagonalize $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ if possible. | 6 | L2 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024

CS3004-1- DESIGN AND ANALYSIS OF ALGORITHMS
(FOR AIDS, AIML, CCE, CSE, ISE)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any **One** full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the properties of an algorithm taking linear search as example.	05	L*2	1	1
b)	For each of the following functions, indicate how much the function value will change if its argument is increased fourfold. i) $\log_2 n$ ii) n iii) n^2 iv) 2^n v) $\sqrt{n}$	05	L3	1	2
2. a)	Explain the general framework for Analysis of an algorithm.	05	L2	1	1
b)	Define 3 asymptotic notations and Determine whether the following assertions are true or false i) $10n^2+9 \in O(n^2)$ ii) $n(n+1)/2 \in \Theta(n^2)$	05	L3	1	2
Unit – II					
3. a)	Write the general plan for analyzing the efficiency of recursive algorithm. Write the pseudo code to find factorial of a number and analyze its efficiency.	06	L3	1	2
b)	Give two brute force algorithms to sort the given list of integers.	04	L2	2	1
4. a)	Illustrate Merge sort algorithm and analyze its efficiency. Sort the following list using Merge sort. 6, 14, 8, 2, 15, 10, 7, 5	06	3	2	2
b)	Show the comparisons the Brute force string matching makes for the pattern $P=0001$ in the text $T=000010001010101$.	04	3	2	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024

CS2102-1 – DATABASE MANAGEMENT SYSTEMS

(For AD, AM, CCE, CSE, ISE,)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any **One** full question from each Unit.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Describe with a neat diagram of Three Schema Architecture. Why do we need mappings among schema levels?	4	L*2	1	1
	b) Design and construct an ER diagram for "Engineering College Management System. Indicate all keys, constraint and assumption that are made.	6	L3	2	2
2.	a) Discuss the main characteristics of database approach.	4	L2	1	1
	b) Explain how domain constraints and referential integrity constraints are specified in relational model.	6	L2	2	1
		Unit – II			
3.	a) Define an Entity attribute. Explain different types of attribute that occur in an E-R diagram model with an example.	5	L2	1	1
	b) Outline the steps to convert the basic E-R model to relational database scheme.	5	L2	2	1
4.	a) For a good database design relation schema should be such that matching attributes must be primary key or foreign key otherwise it leads to spurious tuples". Justify.	5	L3	2	2
	b) Discuss the characteristics of relations.	5	L2	2	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024
MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS
(FOR AM, CCE, CSE, ISE, RI)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I

Marks BT* CO* PO*

1. a) Solve the following system of equations and obtain the general solution.

$$x_1 + 3x_2 - 5x_3 = 4$$

$$x_1 + 4x_2 - 8x_3 = 7$$

$$-3x_1 - 7x_2 + 9x_3 = -6$$

5 L*2 3 1

- b) Solve the following system of equations by Gauss Seidel iteration method. Carry out three iterations, taking (0,0,0) as the initial approximation.

$$2x + y + 6z = 9$$

$$8x + 3y + 2z = 13$$

$$x + 5y + z = 7$$

5 L2 3 1

2. a) By LU decomposition method, factorize the matrix A as LU. Given

$$A = \begin{pmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{pmatrix}$$

5 L2 3 1

- b) For what values of a and b, the system has (i) no solution (ii) unique solution (iii) infinite number of solutions

$$x + 2y + 3z = 6 ; x + 3y + 5z = 9 ; 2x + 5y + az = b$$

5 L3 3 1

Unit – II

3. a) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$

Verify that (i) $tr(B^{-1}AB) = tr(A)$ (ii) $tr(A) = \text{sum of the eigen values of } A$.

5 L2 4 1

- b) Find all eigen values of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$. Find an eigen vector

corresponding to the largest eigen value.

5 L2 4 1

4. a) Determine the largest eigen value and the corresponding eigen

vector of $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ by power method. Take $X^{(0)} = (1 \ 1 \ 1)^T$. Carry out four iterations.

4 L3 4 1

- b) Diagonalize $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ if possible.

6 L2 4 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CSE) (CBCS) Mid Semester Examinations - I, February 2024

CS3005-1 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Describe the roles various status flags and Control flags in Intel 8086 Microprocessor. If the two numbers 1234H and 05A5H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer.	6	L*1	1	1,2
b)	Assume that the address of 1st element in the following array is 2742H: ARR DB 77H, 41H, 27H, 24H What is the contents of DH and DL after the execution of the following instructions: (i) LEA DX, ARR (ii) MOV DX, ARR Justify your answers.	4	L3	1	1,2
2. a)	Suppose that DS=1200H, BX=0100H, SI=0250H, DI = 2024H, SS=1400H, BP=1500H, Starting address of ARRAY=0012H; Determine the physical address of the data. (i) MOV AL, ARRAY[BX] (ii) MOV AL, [BP+SI+200] (iii) MOV DL, ARRAY[BX+DI]	6	L3	1	1,2
b)	Explain the following addressing Modes with example: (i) Direct Addressing Mode (ii) Based Addressing Mode.	4	L2	1	1
Unit – II					
3. a)	Write an Assembly language program to find minimum element in a byte array.	6	L3	1	1,2
b)	Explain the following assembler directives with example: i. ASSUME ii. MACRO	4	L2	1	1
4. a)	Explain the following assembler instructions by giving an example for each. (i)DIV (ii)DAA	6	L2	1	1
b)	Write an Assembly Language Program to calculate LCM of two byte numbers	4	L3	1	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CSE) (CBCS) Mid Semester Examinations - I, February 2024

CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Identify ethical issues that might arise in an professional organization. Also, explain the various types of software applications.	05	L*2	1	1
	b) Explain with a neat diagram spiral software process model.	05	L2	2	2
2.	a) Mention the essential attributes of a good software. Also, explain the structure of a software requirements specification (SRS) document for any software system of your choice.	05	L2	1	2
	b) Explain incremental software process model with a suitable diagram. Also, briefly explain the key differences between waterfall and incremental process model.	05	L2	2	1
Unit – II					
3.	a) Define requirement. Also, discuss various types of non-functional requirements with suitable example for any software system of your choice.	05	L2	1	1
	b) Explain the need of software engineering discipline in an modern era with suitable justifications.	05	L2	1	2
4.	a) List the readers of system requirements. Also, explain user and system requirements along with suitable examples for mental healthcare patient monitoring software (MHC-PMS) system.	05	L2	1	1
	b) Discuss briefly with suitable example the requirement elicitation and analysis activity in requirements engineering (RE) process.	05	L2	1	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024
MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS
 (FOR AM, CCE, CSE, ISE, RI)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Solve the following system of equations and obtain the general solution.
$x_1 + 3x_2 - 5x_3 = 4$
$x_1 + 4x_2 - 8x_3 = 7$
$-3x_1 - 7x_2 + 9x_3 = -6$ | 5 | L*2 | 3 | 1 |
| b) Solve the following system of equations by Gauss Seidel iteration method. Carry out three iterations, taking (0,0,0) as the initial approximation.
$2x + y + 6z = 9$
$8x + 3y + 2z = 13$
$x + 5y + z = 7$ | 5 | L2 | 3 | 1 |
| 2. a) By LU decomposition method, factorize the matrix A as LU. Given
$A = \begin{pmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{pmatrix}$ | 5 | L2 | 3 | 1 |
| b) For what values of a and b, the system has (i) no solution (ii) unique solution (iii) infinite number of solutions
$x + 2y + 3z = 6$; $x + 3y + 5z = 9$; $2x + 5y + az = b$ | 5 | L3 | 3 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$
Verify that (i) $tr(B^{-1}AB) = tr(A)$
(ii) $tr(A) = \text{sum of the eigen values of } A$. | 5 | L2 | 4 | 1 |
| b) Find all eigen values of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$. Find an eigen vector corresponding to the largest eigen value. | 5 | L2 | 4 | 1 |
| 4. a) Determine the largest eigen value and the corresponding eigen vector of $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ by power method. Take $X^{(0)} = (1 \ 1 \ 1)^T$. Carry out four iterations. | 4 | L3 | 4 | 1 |
| b) Diagonalize $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ if possible. | 6 | L2 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - I, February 2024
MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS
 (FOR AM, CCE, CSE, ISE, RI)

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Solve the following system of equations and obtain the general solution.
$x_1 + 3x_2 - 5x_3 = 4$
$x_1 + 4x_2 - 8x_3 = 7$
$-3x_1 - 7x_2 + 9x_3 = -6$ | 5 | L*2 | 3 | 1 |
| b) Solve the following system of equations by Gauss Seidel iteration method. Carry out three iterations, taking (0,0,0) as the initial approximation.
$2x + y + 6z = 9$
$8x + 3y + 2z = 13$
$x + 5y + z = 7$ | 5 | L2 | 3 | 1 |
| 2. a) By LU decomposition method, factorize the matrix A as LU. Given
$A = \begin{pmatrix} 3 & -7 & -2 & 2 \\ -3 & 5 & 1 & 0 \\ 6 & -4 & 0 & -5 \\ -9 & 5 & -5 & 12 \end{pmatrix}$ | 5 | L2 | 3 | 1 |
| b) For what values of a and b, the system has (i) no solution (ii) unique solution (iii) infinite number of solutions
$x + 2y + 3z = 6$; $x + 3y + 5z = 9$; $2x + 5y + az = b$ | 5 | L3 | 3 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$
Verify that (i) $tr(B^{-1}AB) = tr(A)$
(ii) $tr(A) = \text{sum of the eigen values of } A$. | 5 | L2 | 4 | 1 |
| b) Find all eigen values of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$. Find an eigen vector corresponding to the largest eigen value. | 5 | L2 | 4 | 1 |
| 4. a) Determine the largest eigen value and the corresponding eigen vector of $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ by power method. Take $X^{(0)} = (1 \ 1 \ 1)^T$. Carry out four iterations. | 4 | L3 | 4 | 1 |
| b) Diagonalize $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}$ if possible. | 6 | L2 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (AI&DS) (CBCS) Mid Semester Examinations - I, February 2024

MA2006-1 – LINEAR ALGEBRA, STATISTICAL ANALYSIS & COMPUTING

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each** Unit.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Define the rank of a matrix. Reduce the following matrix to row echelon form and hence find its rank. $\begin{bmatrix} 2 & -4 & 3 & -1 & 0 \\ 1 & -2 & -1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$	6	L*3	1	2
	b)	Define the consistency of linear system of equations. Test whether the following equations are consistent or not $2x - 3y + 7z = 5$ $3x + y - 3z = 13$ $2x + 19y - 47z = 32$	4	L1	1	1
2.	a)	Solve the given system of equations by Gauss elimination method. $x + y + z = 6$ $x - y + 2z = 5$ $3x + y + z = 8$	5	L2	1	1
	b)	Use Gauss-Seidel iteration method to solve the following system of linear equations: $20x + y - 2z = 17$ $3x + 20y - z = -18$ $2x - 3y + 20z = 25$ Start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry out three iterations.	5	L2	1	2
		Unit – II				
3.	a)	Find the eigen vectors of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	5	L2	2	1
	b)	Use Rayleigh's power method to find the largest eigen value and eigen vector of the matrix $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ with the initial eigen vector as $[1 \ 1 \ 1]^T$. Carry out 4 iterations.	5	L2	2	1
4.	a)	Define Eigen values and Eigen vector. Find the eigen values of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$	4	L1	2	1
	b)	Find the modal matrix and spectral matrix of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	6	L3	2	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

IV Sem B.E. (CSE) Mid Semester Examinations -I, June 2023

21CS404 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

Duration: 1 Hour

Max. Marks: 20

Note: 1) Answer any One full question from each Unit.
2) Provide necessary diagrams wherever needed

Unit – I

	Marks	BT*	CO*	PO*
1. a) Analyze the essential attributes of a good software				
b) Explain various types of non-functional requirements.	5	L*3	1	1
	5	L2	1	1
2. a) Discuss Waterfall model software process and give any two of its advantages.				
b) Explain the different types of application that may be developed	5	L2	2	1
	5	L3	2	1

Unit – II

3. a) Analyze the areas where standards of acceptable behavior are not bound by laws but by the more tenuous notion of professional responsibility.				
b) Illustrate the Boehm's spiral model of the software process	5	L2	1	1
	5	L2	2	1
4. a) Build the structure of a requirements document	5	L3	1	1
b) Show a process model of the elicitation and analysis process	5	L2	1	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

IV Sem B.E. (CS/IS/AM/CC) Mid Semester Examinations - I, June 2023

21CS401/21IS401/21CC401/21AM401 - LINEAR ALGEBRA AND ITS APPLICATIONS

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit - I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Find the general solution of the system whose augmented matrix is $\left[\begin{array}{cccc c} 1 & -7 & 0 & 6 & 5 \\ -1 & 7 & -4 & 2 & 7 \\ 0 & 0 & 1 & -2 & -3 \end{array} \right]$. | 4 | L*3 | 1 | 1 |
| b) Find LU factorization of $A = \begin{bmatrix} 1 & 3 & -5 & -3 \\ -1 & -5 & 8 & 4 \\ 4 & 2 & -5 & -7 \\ -2 & -4 & 7 & 5 \end{bmatrix}$ | 6 | L3 | 1 | 1 |
| 2. a) (i) Define basic and free variables.
(ii) Which are the basic variables in the system $AX=B$ where $A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix}$ and $B = \begin{bmatrix} 7 \\ -1 \\ 4 \end{bmatrix}$ | 4 | L2 | 1 | 1 |
| (iii) Determine the rank of the coefficient matrix. | 6 | L3 | 1 | 1 |
| b) Solve the following system of equations by inverse method $x+2y+z=4$, $2x+3y-z=3$, $3x+y+2z=3$ | 4 | L2 | 1 | 1 |

Unit - II

- | | | | | |
|---|---|----|---|---|
| 3. a) (i) Define (i) trace and (ii) spectrum of a square matrix A | | | | |
| (ii) If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$, find the trace of AA^T . | 4 | L2 | 2 | 1 |
| b) Diagonalize $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ if possible | 6 | L3 | 2 | 1 |
| 4. a) Determine the eigen values and eigen vectors of $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ | 5 | L3 | 2 | 1 |
| b) Using Cayley Hamilton Theorem find the inverse of $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ | 5 | L3 | 2 | 1 |

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***IV Sem B.E. (CS/IS/AM/CC) Mid Semester Examinations - I, June 2023****21CS402/21IS402/21AM402/21CC402- DESIGN AND ANALYSIS OF ALGORITHMS**

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) With supporting flow diagram explain algorithm design and analysis process	5	L*2	1	1
	b) Design a recursive algorithm to solve $2^n = 2^{n-1} + 2^{n-1}$ and perform mathematical analysis to find the complexity of that algorithm.	5	L3	1	2,3
2.	a) Define three asymptotic notations and express $f(n) = n(n-1)/2$ using three asymptotic notations with proof by its definition	5	L3	1	1
	b) With pseudo code perform complexity analysis of sequential search algorithm for all the three types of efficiencies	5	L3	1	1,2
		Unit – II			
3.	a) What is a brute force design technique? Give brute force string matching algorithm, perform complexity analysis of that algorithms and trace for the example TEXT: JIM_SAW_ME_IN_SHOE_SHOP PATTERN: SHE	5	L3	2	2,3
	b) Apply merge sort to sort the list L,I,S,T,S,O,R,T,I,N,G in alphabetical order. Perform complexity analysis of merge sort	5	L3	2	2,3
4.	a) Give complete Quick sort algorithm and analyze it for complexity. Trace the algorithm for the given unsorted list that to be sorted 5,3,1,9,8,2, 4,7	5	L3	2	2,3
	b) What is an exhaustive search method? For a knap sack problem $n=4$, capacity $m=10$, $w=(4,6,3,5)$ and values $= (30,50,20,10)$, give solution using exhaustive search method.	5	L3	2	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

IV Sem B.E. (CSE) Mid Semester Examinations -I, June 2023

Duration: 1 Hour

21CS403 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Max. Marks: 20

Note: Answer any **One** full question from **each** Unit.

Unit – I

Marks BT* CO* PO*

- | | | | | |
|--|---|-----|---|---|
| 1. a) Explain the general-purpose register and flag register of 8086 architecture. | 6 | L*2 | 1 | 1 |
| b) Describe the difference between the following instruction statements:
(i) MOV AX, 2437H and MOV AX, [2437H]
(ii) MOV AL, [2437H] and MOV AX, [2437H]
Assume that [2436H] = 23H, [2437H] = 56H and [2438H] = 83H | 4 | L3 | 1 | 1 |
| 2. a) Explain the working of following 8086 instructions along with example for each:
i) IMUL
ii) DAA
iii) LEA | 6 | L2 | 1 | 1 |
| b) Indicate the addressing modes of the destination operand and calculate the physical address for the following instructions.
i) ADD [DI], AL
ii) AND [BX+DI], CL
Assume CS=1000h, DS=2000H, SS=3000H, ES=4000H
BX=0FFFH, DI=1A00H, | 4 | L3 | 1 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) Write an ALP program to read a character from keyboard. Display message MICROPROCESSOR if the key pressed is 'M' otherwise, do not display any message. | 6 | L3 | 1 | 1 |
| b) Write the 8086 instruction which will perform the following:
(i) Copy AX to a memory location at offset 245AH in the data segment.
(ii) Rotate the most significant bit of AL into the least significant bit position.
(iii) Mask the Lower four bits of BL.
(iv) Set most significant bit of AX to a 1 but do not affect the other bits. | 4 | L3 | 1 | 1 |
| 4. a) Write an ALP to find maximum value in a byte array and display the maximum value. | 6 | L3 | 1 | 1 |
| b) Define Instruction Queue and explain the use of instruction queue in 8086 architecture. | 4 | L2 | 1 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

IV Sem B.E. (CSE) Mid Semester Examinations -I, June 2023

Duration: 1 Hour

21CS404 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

Max. Marks: 20

Note: 1) Answer any One full question from each Unit.
2) Provide necessary diagrams wherever needed

Unit – I

	Marks	BT*	CO*	PO*
1. a) Analyze the essential attributes of a good software	5	L*3	1	1
b) Explain various types of non-functional requirements.	5	L2	1	1
2. a) Discuss Waterfall model software process and give any two of its advantages.	5	L2	2	1
b) Explain the different types of application that may be developed	5	L3	2	1

Unit – II

3. a) Analyze the areas where standards of acceptable behavior are not bound by laws but by the more tenuous notion of professional responsibility.	5	L2	1	1
b) Illustrate the Boehm's spiral model of the software process	5	L2	2	1
4. a) Build the structure of a requirements document	5	L3	1	1
b) Show a process model of the elicitation and analysis process	5	L2	1	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

IV Sem B.E. (CS/IS/CC/AM) Mid Semester Examinations - II, July 2023

21CS401/21IS401/21CC401/21AM401 - LINEAR ALGEBRA & ITS APPLICATIONS

Duration: 1 Hour

Max. Marks: 20

Note: Answer any One full question from each Unit.

Unit - I

Marks	BT*	CO*	PO*
4	L*2	3	1

1. a) If G and H are subspaces of a vector space V over the field F then prove that $G \cap H$ is also a subspace of V .
b) Find a basis for i) Column space ii) Null space of

$$A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix}$$

2. a) i) Define linearly independent and linearly dependent vectors.
ii) Check whether the set of vectors $\{(1,2,-1), (1,-2,1), (-3,2,-1)\}$ is linearly independent or linearly dependent.
b) Let W be the subspace of R^4 spanned by the set $\{(1,1,1,1), (1,2,3,2), (2,5,6,4), (2,6,8,5)\}$.
(i) Find a basis for W .
(ii) Extend the basis of W to a basis of R^4 .

6 L3 3 2

4 L2 3 2

6 L3 3 2

Unit - II

3. a) Check whether a transformation $T: R^2 \rightarrow R^2$ defined by $T(x,y) = (3x+2y, 3x-4y)$ is a linear transformation or not.
b) i) State Rank-Nullity theorem.
ii) The linear transformation $T: R^3 \rightarrow R^3$ defined by $T(x,y,z) = (x+2y-z, y+z, x+y-2z)$. Find the nullity of T and rank of T .

4 L2 4 2

6 L3 4 2

4 L2 4 1

4. a) i) Define null space of a linear transformation $T: V \rightarrow W$.
ii) Prove that null space of T is a subspace of V .
b) A linear transformation defined by $T(x,y,z) = (3x+2y-4z, x-5y+3z)$. Find the matrix of linear transformation relative to the bases $\beta_1 = \{(1,1,1), (1,1,0), (1,0,0)\}$ and $\beta_2 = \{(1,3), (2,5)\}$.

6 L3 4 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Fourth Semester B.E. (CSE) (Credit System) Degree Examinations
 August 2023

21CS403 – MICROPROCESSOR AND EMBEDDED SYSTEMS

Max. Marks: 100

Duration: 3 Hours

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Briefly Explain General purpose registers and flag register with respect to Intel 8086. | 8 | L*2 | 1 | 1,2 |
| b) Explain the working of following 8086 instructions along with example for each:
(i) AAM (ii) DAA | 4 | L2 | 1 | 1,2 |
| c) Assume that BX is 1020H and the content of 101FH = 08H; content of 1020H = 3FH; Content of 1021H = 6EH. Which value is stored in CX register after execution of the statement MOV CX, [BX] Justify your answer. | 4 | L3 | 1 | 1,2 |
| d) Write an ALP to calculate LCM of two byte numbers. | 4 | L3 | 1 | 1,2 |
| 2. a) What is the role of BIU and EU of Intel 8086 microprocessor? Explain BIU in detail. | 8 | L2 | 1 | 1,2 |
| b) Explain the following addressing Modes with instruction example:
(i) Direct Addressing Mode (ii) Register Relative Addressing mode (ii) Based Indexed Addressing Mode. | 6 | L2 | 1 | 1,2 |
| c) Write an ALP to reverse a given string and check whether it is a palindrome or not. | 6 | L3 | 1 | 1,2 |
| 3. a) How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example. | 8 | L2 | 1 | 1,2 |
| b) Briefly explain the following Assembler directives used in Intel 8086 microprocessor:
(i) DD (ii) SEGMENT (iii) EQU (iv) ASSUME | 8 | L1 | 1 | 1,2 |
| c) Assume that the address of 1st element in the following array is 2742H:
ARR DB 77H, 41H, 27H, 24H
What is the contents of DH and DL after the execution of following instructions:
(i) LEA DX, ARR (ii) MOV DX, ARR
Justify your answers. | 4 | L3 | 1 | 1,2 |

Unit – II

- | | | | | |
|---|----|----|---|-------|
| a) Write an embedded Program to display a "FIREHELP" using right to left scrolling effect on a 7-segment display. | 6 | L3 | 4 | 1,2,3 |
| b) Discuss Embedded system hardware design. | 10 | L1 | 3 | 1,2 |
| c) Discuss Embedded system hardware design and a Procedure. | 4 | L2 | 2 | 1,2,3 |
| | | | 3 | 1,2 |

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (CSE/ISE/AIML/CCE) (Credit System) Degree Examination

Make up Examinations - September 2023

21CS402/21IS402/21AM402/21CC402 - DESIGN AND ANALYSIS OF ALGORITHMS

Duration: 3 Hours

Max. M

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	C
1.	a) Write an Algorithm to check whether all the elements in a given array are distinct. Derive its time efficiency.	6	L*3	
	b) Discuss Brute Force string matching Algorithm. What is the efficiency of this Algorithm in worst case?	6	L3	
	c) Write an Algorithm to sort n numbers using Merge sort. Discuss its efficiency in worst case.	8	L3	
2.	a) Design an Algorithm to search an element using sequential search. Discuss its efficiency in best, worst and average cases.	6	L3	
	b) Show the complexity of selection sort by writing and analyzing an Algorithm.	6	L4	
	c) Apply Quick sort to sort the following numbers in alphabetical order. M,E,R,G,E,S,O,R,T Discuss the time complexity of quick sort in all three cases.	8	L3	
3.	a) Discuss Big Oh and Big theta asymptotic notations with suitable example.	6	L2	
	b) Write recursive Algorithm to find the number of binary digits in positive decimal number n. Show its complexity.	6	L3	
	c) Discuss general Divide and Conquer procedure. Write an Algorithm to search an element in an array using Binary search.	8	L2	

Unit – II

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

IV Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2025

MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS
 (For AM, CB, CC, CS, IS, RI)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions

Note: Answer all Four questions in the Answer Book. Each question carries equal marks.

- In three dimensional Euclidean space inverse of $(-2, -2, 0)$ with respect to vector addition is
 A) $(2, 2, 2)$ B) $(-1, -1, 0)$
 C) $(-2, -2, 0)$ D) $(2, 2, 0)$
- If a basis of a vector space has 3 elements, then dimension of the vector space is
 A) 2 B) 1
 C) 3 D) 0
- Let $T: R^3 \rightarrow R^4$ be the linear transformation defined by
 $T(x_1, x_2, x_3) = (x_1, 0, x_2, x_3)$. Then basis for null space of T is
 A) $(1, 0, 0)$ B) $(0, 0, 1)$
 C) $(0, 1, 0, 0)$ D) $(1, 1, 0, 0)$
- Which of the following statement is not true if $T: U \rightarrow V$ is a linear transformation
 A) V is always the range of T B) Null space of T is a subspace of U
 C) Range space of T is a subspace of V D) $T(0) = 0$

Part – B: Descriptive Answer Questions

Note: Answer any One full question from each Unit.

Unit – I

- a) i) Define a subspace of a vector space V over a field F.
 ii) If G and H are subspaces of a vector space V over F then prove that $G \cap H$ is also a subspace of V.
 b) Define (i) Row space (ii) Null space of a matrix.

Check whether the vector $\begin{pmatrix} 3 \\ 3 \\ -4 \end{pmatrix}$ is in the column space of the

$$A = \begin{pmatrix} 1 & -3 & 4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{pmatrix}$$

matrix

- a) i) Define Linearly dependent and Linearly independent of a set of vectors.

ii) Express $u = (1, -2, 5)$ in R^3 as a linear combination of the vectors
 $u_1 = (1, 1, 1)$ $u_2 = (1, 2, 3)$ $u_3 = (2, -1, 1)$ $\Rightarrow (-6, 3, 2)$

- b) Find the dimension of the subspace W of R^3 spanned by the vectors $\{(2, 4, 2), (1, -1, 0), (1, 2, 1), (0, 3, 1)\}$. Extend the basis of W to a basis of R^3 .

Unit – II

- a) (i) State Rank-Nullity theorem.

(ii) Let a linear transformation $T: R^3 \rightarrow R^3$ be defined by

$T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Find the null space of T and range space of T. Also find dimension of the null space of T and dimension of range space of T.

Marks BT* CO* PO*

4 L*1 3 1

4 L2 3 2

5 L1 3 2

3 L2 3 1

5 L3 4 2

P.T.O.

- b) Check whether a transformation $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (x+y, x)$ is a linear transformation or not.
4. a) Find the matrix of the linear transformation $T: R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - y + 2z, 3x + y)$ with respect to bases $B = \{(1, 1, 1), (1, 2, 3), (1, 0, 0)\}$ and $B' = \{(1, 1), (1, -1)\}$.
- b) Let V and W be vector spaces over the field F . Let T and U be linear transformation from V into W . Then prove that the function $(T+U)$ defined by $(T+U)\alpha = T(\alpha) + U(\alpha)$ is a linear transformation from V into W . Also prove, if c is any element of F , the function (cT) defined by $(cT)(\alpha) = c(T\alpha)$ is a linear transformation from V into W .

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

3 L2 4 1

$$\begin{bmatrix} 3 & -1 \\ 5 & 2 \end{bmatrix}$$

4 2

3 L2 4 1

$T(1, 1, 1)$

$1 - 1 + 2 = 2$

$3 + 1 = 4$

$1, 2, 3$

$1 - 2 + 2(3)$

$3 + 2$

$1, 0, 0$

$1 - 0 + 0$

$(2, 4) = a_{11}(1, 1) + a_{21}(1, -1)$

$a_{11} + a_{21} = 2$

$a_{11} - a_{21} = -2$

$a_{11} + a_{11} + 1 = 2$

$6a_{11} = 3$

$a_{21} = -1$

$a_{11} = 1$

$2a_{11}$

$2a_{11}$

$$\begin{bmatrix} 3 & 5 & 2 \\ -1 & 0 & -1 \end{bmatrix}$$

$a_{11} a_{11}$

a_{21}

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations-II, April 2025
CS2103-1 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT
 (CS, CB)

Duration: 1 Hour

Max. Marks: 20

Part – A: Multiple Choice Questions (1*4 = 4 marks)

*Note: Answer all **Four** questions in the **Answer Book**. Each question carries equal marks.*

1. Which of the following diagrams is used to model the vocabulary of a system?

A) Object Diagram

C) Class diagram

B) Activity Diagram

D) Interaction Diagram
2. Composition is a stronger form of which of the following?

A) Aggregation

C) Encapsulation

B) Inheritance

D) Generalization
3. Select the option that suits the Manifesto for Agile Software Development

A) Individuals and interactions

C) Customer collaboration

B) Working software

D) All the above mentioned choices are correct
4. What encapsulates both data and data manipulation functions?

A) Object

C) Super Class

B) Class

D) Sub Class

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

*Note: Answer any **One** full question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a neat diagram explain Context model of MHC-PMS.	4	L2	3	2
b)	Draw and explain the Sequence diagram for View Patient Information.	4	L2	3	2
2. a)	Explain the Activity model of an Insulin's pump operation.	4	L2	3	2
b)	Outline the Use case description for Report Weather.	4	L2	3	2
Unit – II					
3. a)	Explain various principles of Agile methods.	4	L2	3	2
b)	Explain Test Driven Development Process with diagram.	4	L2	4	2
4. a)	Explain any four Extreme Programming practices.	4	L2	3	2
b)	Explain the Model of software testing process with diagram.	4	L2	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CSE) Mid Semester Examinations - II, APRIL 2025

CS3005-1 – MICRPROCESSOR AND EMBEDDED SYSTEM

Max. Marks: 20

Duration: 1 Hour

Part – A: Multiple Choice Questions (1*4 = 4 marks)

Note: Answer all Four questions in the Answer Book. Each question carries equal marks.

1. Which of the following is NOT an advantage of modular programming?
A) Code reusability
B) Easier debugging
C) Increased execution time
D) Better program organization
2. Which interrupt is used to set video mode in 8086?
A) INT 10H, AH = 00H
B) INT 16H, AH = 00H
C) INT 21H, AH = 01H
D) INT 13H
3. Which register in ARM holds the flags indicating the status of the processor?
A) General-purpose register
B) Program Counter (PC)
C) Stack Pointer (SP)
D) Program Status Register (PSR)
4. What is the size of the general-purpose registers in ARM processors?
A) 8 bits
B) 16 bits
C) 32 bits
D) 64 bits

Part – B: Descriptive Answer Questions (2*8 = 16 marks)

Note: Answer any One full question from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Compare and contrast RISC and CISC architectures | 4 | L2 | 3 | 1 |
| b) Write the procedure for reading a 2-digit hexa decimal number from the standard input device. | 4 | L3 | 3 | 1 |
| 2. a) Discuss ARM design Philosophy. | 4 | L2 | 3 | 1 |
| b) Write an ALP to read an alphanumeric character and display its equivalent ASCII code at the center of the screen | 4 | L3 | 3 | 1 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 3. a) Write an embedded C program to control the speed of rotation of a stepper motor using potentiometer. | 4 | L3 | 4 | 1 |
| b) What is Arduino communication and explain the types of Arduino communication. | 4 | L2 | 4 | 1 |
| 4. a) With a code snippet, explain pinMode(), digitalWrite(), and analogRead() functions. | 4 | L2 | 4 | 1 |
| b) Write an Embedded C program to scroll the word 'FIRE' from left to right on a 7-segment display. | 4 | L3 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV SEM B.Tech (CBCS) Mid Semester Examinations - II, April 2025
CS3004-1 – DESIGN AND ANALYSIS OF ALGORITHMS
 (For AD, AM, CB, CC, CS, IS)

Duration: 1 Hour

Max. Marks: 20

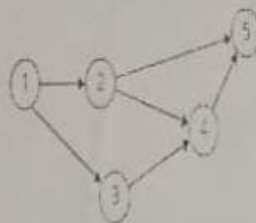
Part – A: Multiple Choice Questions (1*4 = 4 marks)*Note: Answer all Four questions in the Answer Book. Each question carries equal marks.*

- Which of the following is a necessary condition for a graph to have a topological sorting?
 A) The graph must be a tree
 C) The graph must be a complete graph
 B) The graph must be a DAG
 D) The graph must be a bipartite graph
- What type of data structure is used in Depth-First Search (DFS)?
 A) Queue
 C) Priority queue
 B) Heap
 D) Stack
- What is the time complexity of distribution counting sort?
 A) $O(n^2)$
 C) $O(2^n)$
 B) $O(n \log n)$
 D) $O(n)$
- In dynamic programming, the solution to a problem is built up from;
 A) A series of overlapping sub problems
 C) A greedy approach
 B) A single recursive call
 D) A divide and conquer method

Part – B: Descriptive Answer Questions (2*8 = 16 marks)*Note: Answer any One full question from each Unit.***Unit – I**

Marks BT* CO* PO*

- a) Outline Insertion sort algorithm and show the sorting of list E, X, A, M, P, L, E in alphabetical order using insertion sort method. 4 L3 3 2
 b) Apply heap sort algorithm to sort the list 90, 89, 56, 72, 67, 12, 21, 34, 45. 4 L3 3 2
- a) Define AVL tree and construct AVL tree for the data {2, 9, 3, 12, 15, 8, 11} 4 L3 3 2
 b) Define topological sorting. Illustrate the topological sorting using Source Removal method for the following graph.



4 L3 3 2

Unit – II

- a) Write an algorithm to sort list of integers using Comparison Counting and compare its time complexity with the time complexity of selection sort and Bubble sort. 4 L3 4 2
 b) Summarize the steps of Horspool string matching algorithm and search for the pattern TIME in the text A STITCH IN TIME SAVES NINE. 4 L3 4 2

P.T.O

CS3004-1

MSE-II APRIL 2025

4. a) Apply All Pair Shortest path algorithm, for the following graph.



- b) Write an algorithm for Breadth-First Search (BFS). List any two facts that differentiate BFS from DFS.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

4	L3	4	2
4	L3	3	2

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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
IV Sem B.Tech (CBCS) Mid Semester Examinations - II, APRIL 2025

CS2102-1 – DATABASE MANAGEMENT SYSTEMS
CS, IS, AM, AD, CC

Duration: 1 Hour

Max. Marks: 15

Part – A: Multiple Choice Questions (1*3 = 3 marks)

*Note: Answer all **Three** questions in the **Answer Book**. Each question carries equal marks.*

1. What is the result of the query **SELECT COUNT (DISTINCT Salary) FROM EMPLOYEE;**?
A) Total number of employees
B) Total number of salaries greater than a specified value
C) Total number of unique salaries
D) Average salary of employees
2. What type of dependency does 2NF aim to eliminate?
A) Partial dependency
B) Referential dependency
C) Transitive dependency
D) Candidate key dependency
3. In a relation, if attribute B is functionally dependent on attribute A, which of the following is true?
A) Each value of B is associated with one value of A
B) Each value of A is associated with one value of B
C) There is no dependency between A and B
D) A and B must be the same attribute

Part – B: Descriptive Answer Questions (2*6 = 12 marks)

*Note: Answer any **One** full question from **each Unit**.*

Unit – I

Marks BT* CO* PO*

1. a) Consider the following schema and write the SQL queries for the following:
Suppliers (sid: integer, **sname**: string, **address**: string)
Parts (pid: integer, **pname**: string, **color**: string)
Catalog (sid: integer, pid: integer, **cost**: double)
i. Find the sids of suppliers who supply some red or green part
ii. Find the sids of suppliers who supply only red parts
iii. Find the pnames of parts for which there is some supplier
iv. Find the sids of suppliers who supply some red part or are located at 221 Packer Street. (solution should be in Nested Query)

6 L3 3 2

- a) Consider the following schema and write the SQL queries for the following:
Movie (movie_id, title, release_year, genre, director_id, rating)
Director (director_id, name, birth_year, nationality)
Actor (actor_id, name, birth_year, nationality)
Movie_Actor (movie_id, actor_id, role)
Review (rev_id, user_id, movie_id, rev_text, rating, rev_date)
User (user_id, username, email, age)
i. Find the names of all actors who have acted in at least one movie released after 2015.
ii. Find the directors who have directed more than 5 movies.
iii. Get the average rating of all movies directed by 'Christopher Nolan'.
iv. Find movies that have a higher rating than the average rating of movies released in the same genre. (solution should be in Nested Query)

6 L3 3 2

CS2102-1

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Unit - II

3. a) Consider the two sets of functional dependencies
 $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$ Are they
 equivalent? 4 L3 4 2
- b) What is Normalization? Explain the first Normal form with example. 2 L2 4 1
4. a) Suppose a relational schema R (A B C D E F G H I J) and set of
 functional dependencies
 $X: \{ \{A, B\} \rightarrow C, A \rightarrow \{D, E\}, B \rightarrow F, F \rightarrow \{G, H\}, D \rightarrow \{I, J\} \}$
 What is candidate key of R? Check out whether the relation is in 3NF
 or not? If not decompose it in 3NF. 4 L3 4 2
- b) Explain any two Informal Design Guidelines for Relation Schemas. 2 L2 4 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

4th Sem
 MSE II

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

IV Sem B.E. (CS/IS/CC/AM) Mid Semester Examinations - II, July 2023

21CS401/21IS401/21CC401/21AM401 - LINEAR ALGEBRA & ITS APPLICATIONS

Duration: 1 Hour

Max. Marks: 20

Note: Answer any **One** full question from **each Unit**.

Unit – I

Question	Marks	BT*	CO*	PO*
1. a) If G and H are subspaces of a vector space V over the field F then prove that $G \cap H$ is also a subspace of V .	4	L*2	3	1
b) Find a basis for i) Column space ii) Null space of $A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix}$	6	L3	3	2
2. a) i) Define linearly independent and linearly dependent vectors. ii) Check whether the set of vectors $\{(1, 2, -1), (1, -2, 1), (-3, 2, -1)\}$ is linearly independent or linearly dependent.	4	L2	3	2
b) Let W be the subspace of R^4 spanned by the set $\{(1, 1, 1, 1), (1, 2, 3, 2), (2, 5, 6, 4), (2, 6, 8, 5)\}$. (i) Find a basis for W . (ii) Extend the basis of W to a basis of R^4 .	6	L3	3	2

Unit – II

Question	Marks	BT*	CO*	PO*
3. a) Check whether a transformation $T : R^2 \rightarrow R^2$ defined by $T(x, y) = (3x + 2y, 3x - 4y)$ is a linear transformation or not.	4	L2	4	2
b) i) State Rank-Nullity theorem. ii) The linear transformation $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Find the nullity and rank of T .	6	L3	4	2
4. a) i) Define null space of a linear transformation $T : V \rightarrow W$. ii) Prove that null space of T is a subspace of V .	4	L2	4	1
b) A linear transformation defined by $T(x, y, z) = (3x + 2y - 4z, x - 5y + 3z)$. Find the matrix of linear transformation relative to the bases. $\beta_1 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $\beta_2 = \{(1, 3), (2, 5)\}$.	6	L3	4	2

Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

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IV Sem B.E. (CSE/ISE/AIM/CCE) Mid Semester Examinations - II, July 2023

20CS402 / 21IS402 / 21AM402 / 21CC402 – DESIGN AND ANALYSIS OF ALGORITHMS

Duration: 1 Hour

Max. Marks: 20

Note: Answer any **One** full question from **each Unit**.

Unit – I

Question	Marks	BT*	CO*	PO*
1. a) Traverse the graph below using DFS and BFS. Also write the DFS and BFS tree for the same.	5	L*2	3	1,2
b) Write insertion sort algorithm and trace it for the following input: ALGORITHMS	5	L2	3	1,2
2. a) Write the algorithm for BottomUp Heap construction. Apply Heapsort for the following input. Show detailed construction of heap for every step. <i>Input:</i> 28 15 45 6 17 32 3 12	5	L2	3	1,2
b) Explain the two methods for finding the topological order for any graph. Apply any of the methods to write the topological sort order for following graph.	5	L2	3	2,3

Unit – II

Question	Marks	BT*	CO*	PO*
3. a) Construct AVL Tree for the input given below. Explain the types of rotations used in the construction. 12, 56, 89, 16, 25, 100, 75, 55, 95, 50	5	L2	3	2,3
b) Write Horspool's String Matching Algorithm and apply the same for the following input: <i>Text</i> : NITTE UNIVERSITY <i>Pattern</i> : SIT	5	L2	4	2,3
4. a) Write Floyd's Algorithm and apply the same for the distance matrix given below: (Note: inf stands for infinity) 0 2 4 3 5 0 inf 3 5 inf 0 3 inf 1 4 0	5	L2	4	2,3
b) Construct 2-3 Tree for the input given below. INFORMATIVE	5	L2	3	1,2

Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

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IV Sem B.E. (CSE) Mid Semester Examinations -II, July 2023 21CS403 – MICROPROCESSOR AND EMBEDDED SYSTEM

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each** Unit.*

Unit – I

Question	Marks	BT*	CO*	PO*
1. a) Write the procedure to convert two ASCII values to the equivalent hexadecimal. Write appropriate comments.	4	L*2	2	1
b) Develop an assembly language program to extract the hour from the system time and display in decimal at row number 8 and column number 20 in the format: HOUR: 10 where 10 is the current hour extracted from the system time. Write appropriate comments.	6	L3	2	2
2. a) Explain the interrupts required to perform the following tasks by specifying the necessary parameters involved: i. Buffered keyboard input ii. Set system date iii. Get cursor position iv. Create sub directory	4	L2	2	1
b) Develop an assembly language program to read a string from the keyboard and write the same to a file. Display the appropriate error messages. Write appropriate comments.	6	L3	2	2

Unit – II

Question	Marks	BT*	CO*	PO*
3. a) Bring out the various differences between RISC and CISC machines.	5	L2	3	1
b) Develop an embedded C program to display a message using scrolling effect on LCD for a suitable period of time in both directions. Do not use the built-in function for scrolling.	5	L3	4	2
4. a) Explain the embedded system software in detail.	5	L2	3	1
b) Develop an embedded C program to perform up-down counter on a 7-segment display using switches.	5	L3	4	2

Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

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IV Sem B.E. (CSE) Mid Semester Examinations -II, July 2023 21CS404 – SOFTWARE ENGINEERING AND PROJECT MANAGEMENT

Duration: 1 Hour

Max. Marks: 20

*Note: 1) Answer any **One** full question from **each Unit**.
2) Provide necessary diagrams wherever needed*

Unit – I

Question	Marks	BT*	CO*	PO*
1. a) With neat diagram explain the organization of the Model-View-Controller.	5	L*2	2	1,2
b) Explain the context model with a neat diagram.	5	L2	2	1
2. a) Draw and explain the sequence diagram for describing data collection for Weather Information System.	5	L2	3	1
b) Develop an architecture for a language processing system.	5	L3	2	1

Unit – II

Question	Marks	BT*	CO*	PO*
3. a) Construct a state diagram of a microwave oven system.	5	L3	2	1,2
b) List and explain the principles of the agile method.	5	L2	3	1
4. a) Briefly explain the five stages of the object-oriented design process.	5	L2	3	1
b) Explain the repository model with a neat block diagram.	5	L2	2	1

Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

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IV Sem B.E. (CS/IS/AM/CC) Mid Semester Examinations - II, June 2022

20CS401 / 20IS401 / 20AM401/ 20CC401 – LINEAR ALGEBRA & PROBABILITY THEORY

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I

Question	Marks	BTL*	CO*	PO*
1. a) Check whether a transformation $T : R^2 \rightarrow R^3$ defined by $T(x, y) = (3x + y, 5x + 7y, x + 3y)$ is a linear transformation or not.	4	L*2	3	1
b) i) State Rank-Nullity theorem. ii) Let a linear transformation $T : R^3 \rightarrow R^3$ be defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$. Find the null space of T and rank of T.	6	L3	3	1
2. a) Find the matrix of the linear transformation $T : R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x - y + 2z, 3x + y)$ with respect to bases $B = \{(1, 1, 1), (1, 2, 3), (1, 0, 0)\}$ and $B' = \{(1, 1), (1, -1)\}$.	4	L2	3	
b) Construct an orthonormal basis for a subspace W of R^3 with a basis $\{x_1, x_2, x_3\}$ where $x_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $x_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $x_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ of W, using Gram_schmidt process.	6	L3	3	1

Unit – II

Question	Marks	BTL*	CO*	PO*														
3. a) Find a least square solution of $AX = B$ where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.	4	L2	3	1														
b) i) The pdf $p(x)$ of a variate X is given by the following table <table><tr><td>X</td><td>0</td><td>1</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>p(x)</td><td>k</td><td>3k</td><td>5k</td><td>9k</td><td>10k</td><td>12k</td></tr></table> For what value of k, does this represent a valid probability distribution? Also find $P(X > 3)$ and $E[X]$. ii) If $f(x) = kx^3, 0 < x < 1$ $= 0$, elsewhere is a pdf of the random variable X, find (i) the value of k (ii) $P(X > 0.7)$ (iii) $P\left(\frac{1/4 < X < 3/4}{X > 1/2}\right)$	X	0	1	3	4	5	6	p(x)	k	3k	5k	9k	10k	12k	3+3	L2	4	1
X	0	1	3	4	5	6												
p(x)	k	3k	5k	9k	10k	12k												
4. a) If X is a random variable, prove that (i) $V(X) = E(X^2) - [E(X)]^2$. (ii) $V(aX + b) = a^2V(X)$ where a and b are constants.	4	L2	4	1														
b) i) State Bayes' theorem. ii) A business man goes to hotels X, Y, Z 30%, 45% and 25% of the time respectively. It is known that 5%, 4%, 6% of the rooms in hotels X, Y, Z respectively have faulty plumbing. Determine the probability (i) that the business man goes to hotel with faulty plumbing (ii) that business man's room having faulty plumbing is assigned to hotel Y?	6	L3	4	1														

Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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NMAM INSTITUTE OF TECHNOLOGY, NITTE

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IV Sem B.E. (CS/IS/AM/CC) Mid Semester Examinations – II, June 2022 20CS402 / 20IS402 / 20AM402 / 20CC402 – DESIGN AND ANALYSIS OF ALGORITHMS

Duration: 1 Hour

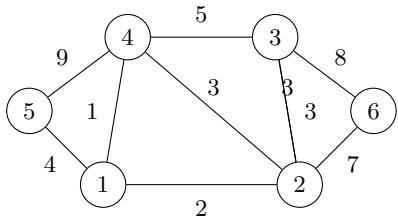
Max. Marks: 20

Note: Answer any **One** full question from **each Unit**.

Unit – I

Question	Marks	BT*	CO*	PO*
1. a) Develop an algorithm to perform Insertion Sort and analyze the best and worst case efficiencies.	4	L*4	3	2
b) Find the solution for following problem using topological sorting method. Consider a set of 5 courses{C1,C2,C3,C4,C5}, a part time student has to take in some degree program. The courses can be taken in any order as long as the following course prerequisites are met: C1 and C2 have no prerequisites, C3 requires C1 and C2, C4 requires C3 and C5 requires C3 and C4. The student can take only one course per term.	3	L3	3	2
c) Construct 2-3 tree for the data C,O,M,P,U,T,I,N,G.	3	L3	3	2
2. a) Explain Horspool's algorithm and apply the algorithm to find the pattern "MONEY" in the text "A FOOL AND HIS MONEY ARE PARTED SOON".	4	L3	3	2
b) Apply closed hashing to distribute the keys 10,23,14,19,17,32,45 using the function $H(k) = K(\text{mod}11)$.	2	L3	3	2
c) Construct heap and sort using array representation 2,9,8,14,22,16,7	4	L3	3	2

Unit – II

Question	Marks	BT*	CO*	PO*												
3. a) Outline the pseudocode for Prim's algorithm to find the minimum spanning tree. Apply the Prim's algorithm to find the minimum spanning tree for the following graph 	6	L3	4	2												
b) Write a pseudocode for Bellman Ford algorithm to find the single source shortest path.	4	L2	4	1												
4. a) Construct a Huffman code for the following data: <table border="1" data-bbox="150 1803 676 1870"><tr><td>Character</td><td>A</td><td>B</td><td>C</td><td>D</td><td>–</td></tr><tr><td>Probability</td><td>0.4</td><td>0.1</td><td>0.2</td><td>0.15</td><td>0.15</td></tr></table> Encode the text ABACABAD using the code of obtained. Decode the text whose encoding is 100010111001010	Character	A	B	C	D	–	Probability	0.4	0.1	0.2	0.15	0.15	6	L3	4	1
Character	A	B	C	D	–											
Probability	0.4	0.1	0.2	0.15	0.15											
b) Write the pseudocode for single source shortest path in DAG.	4	L2														

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
